Predictive Risk and Recording Decisions in Partially Observed Neural Networks

Jiayi Du[1*†] and Shuaizhi Cheng[2†]

[1*]Stiftung Louisenlund, Germany.
[2]Department of Computing, Harbin Institute of Technology, China.

*Corresponding author(s). E-mail(s): jiayi.du@mail.louisenlund.de;
†These authors contributed equally to this work.

Abstract

Neural interfaces in biohybrid systems use stimulation and recording to interact with living networks. Under a limited recording budget, choosing where to record becomes part of predicting a specified stimulus response. Goal-oriented design ranks recording sets by estimated prediction error. We asked whether averaging this local risk over a parameter prior improves recording decisions, and which recordings make a specified response predictable when connectivity is known but neuronal gains are uncertain.

We first compared nominal-parameter and prior-averaged risk criteria in simulated 12-unit networks with uncertain neuronal gains and biases, with the measurement budget and maximum a posteriori (MAP) estimator held fixed. Across 40 independent networks, prior averaging changed mean squared response error (MSE) by -3.74% for a single-response objective and +2.69% for a shared three-response objective; both paired network-cluster intervals included zero. The two criteria chose identical recordings in 68% and 65% of designs, and a separately specified risk-forecast study did not establish a better forecast.

In 100-unit rank-two rate networks with all sites available, selectors under both criteria always recorded the stimulation site and included all three readouts in 30 and 38 of 40 networks; we call these four sites the endpoints. We therefore controlled which endpoints occupied five calibration slots; the target stimulus response was withheld from fitting. Recording all four endpoints reduced held-out response MSE by 96.44% relative to five non-endpoint recordings in 40 networks, and by 98.13% in an independent 40-network validation (95% CI, -99.19% to -93.94%). The benefit was specific to the intended response; whole-network activity under other inputs showed no clear improvement. In 40 further networks, endpoint recording narrowed numerically feasible response ranges (not calibrated intervals; mean squared width, -99.08%), and lower recording precision weakened the benefit. In the first cohort, neither criterion showed a clear benefit over this endpoint baseline.

An exact factorization of the finite response into input, recurrent-propagation and readout terms localized the strongest error contraction to the input and readout terms. Oracle controls that revealed true gains indicated that four recurrent gain combinations add predictive value beyond endpoint gains (independent cohort, -68.41%; 95% CI, -94.86% to -10.68%; the interval adjusted for three comparisons included zero). With the endpoints retained, a fifth recording chosen for these combinations narrowed a feasible-range width along a direction fixed from the prior by 27.15% relative to a random fifth site. In 80 further networks, this choice reduced combination-estimation MSE by 31.07% (secondary comparison; 95% CI, -42.35% to -19.81%), but neither this choice nor a response-directed fifth recording clearly differed from a random fifth site in response MSE.

Endpoint benefits persisted under rank-six augmentation and mild connectivity or initial-state errors, with less certainty in independently generated Gaussian rank-six and positive-gain networks. These simulations indicate that endpoint coverage is an informative baseline for evaluating recording criteria, and that better estimation of response-relevant parameters is distinct from better response prediction. Applicability to living networks and mechanical outputs remains to be tested.

1 Introduction

A recording set should be judged by how well its data support the intended prediction. In a neural interface, that prediction may concern the response at a few output sites following a planned stimulus. Cultured networks support decoding of patterned inputs [1] and closed-loop interaction through small electrode sets [2]. Biohybrid systems also couple neural or muscle stimulation to mechanical output [3–6]. These applications motivate choosing recordings for a specified response. Here we study the neural prediction problem in simulated rate networks with known connectivity, uncertain neuronal parameters and a limited recording budget.

Connectivity constrains neural dynamics but does not fix all neuronal properties. Different circuit parameters can produce similar activity [7], and uncertain parameters can coexist with well-constrained predictions [8]. Connectome-constrained models use partial activity recordings to constrain uncertain physiology [9], while models combining anatomical structure and task optimisation can predict visual responses [10]. For a planned stimulus, the stimulation and readout sites are natural candidates for recording because they determine how the signal enters and is observed. We refer to these sites as the endpoints. Their value can be tested by changing which endpoints occupy a fixed number of calibration slots and evaluating a stimulus response withheld from fitting.

Recording-site selection has an established literature. Brain-computer interfaces select channels for decoding [11], and high-density probes offer more candidate sites than can be read out at once [12]. In nonlinear networks, structural observability and practical state-estimation accuracy are distinct [13]; greedy sensor selection has guarantees for suitable objectives [14]. For the present task, the quantity to assess is the error of the fitted model's response to an unobserved stimulus. Perturbation outcomes are increasingly used to evaluate neural models [15]. We fix the stimulation protocol and choose the calibration recording set; experimental design of the stimulation input addresses a related question [16, 17].

Bayesian experimental design values observations by the expected utility of the resulting inference [18, 19], with goal-oriented designs focusing on a specified quantity [20]. Gaussian posterior approximations make covariance-based design tractable [21], while sampling-based approaches address nonlinear inference more fully [22–25]. We compare a local response-risk criterion evaluated at nominal parameters with its average over a prior ensemble. These methods provide a starting point for selecting recordings. Their benefit depends on which task-relevant information the observations supply.

We began by comparing nominal and prior-averaged recording criteria, then examined the role of endpoint coverage in the selected sets. Independent networks confirmed a large benefit from joint endpoint recording. We test how those observations constrain the responses compatible with the calibration data and how this changes with recording precision. A factorization into input, recurrent propagation and readout identifies candidate parameter combinations; information interventions test their predictive value after endpoint gains are known. A fifth-recording experiment then asks whether observations can acquire some of the remaining internal information. Structural perturbations examine the scope of the allocation result. The original criterion comparisons retain their prespecified status and are reported alongside these focused follow-ups.

2 Results

2.1 Predicting a stimulus response from limited calibration recordings

The endpoint and mechanism experiments use 100-unit rate networks with known connectivity and uncertain neuronal gains. Calibration records activity at a limited set of sites under known inputs. The target is the time-integrated response at three fixed readouts to a separate stimulus, relative to an unstimulated control. Stimulation and readout sites are chosen before generating the hidden physiology and observation noise. The response to be predicted is withheld from fitting.

Each allocation comparison holds the recording budget, calibration inputs, noise and fitting rule fixed within network. Parameters are fitted to the selected observations and the model then predicts the stimulus response by nonlinear simulation. The initial endpoint experiment follows a comparison of nominal and prior-averaged recording criteria; independent cohorts test the allocation benefit and its information mechanism. The original criterion studies and their frozen primary results are summarized below and reported in full in the supplement.

2.2 Joint endpoint recording improves response prediction

When all sites were available and five were recorded, nominal and ensemble shared-objective selectors reduced MSE relative to random recording by 95.95% and 88.17%, respectively. Inspection of the selected sets suggested a concrete source of this benefit: both selectors always recorded the stimulation site, and they included all three readouts in 30 and 38 of the 40 networks. We therefore introduced a simple endpoint baseline and controlled which endpoints occupied the same five recording slots.

Four allocations recorded neither endpoint group, the stimulation site only, the three readouts only, or both groups. Unfilled slots were sampled from the remaining neurons using shared random permutations, with three allocation repeats. All four allocations used the same hidden networks, calibration data, recording count and original estimator. In the first 40 networks, recording both groups changed MSE by -96.44% (95% CI -98.38% to -92.52%) relative to recording neither, with improvement in 38 networks. Relative to stimulation-only and readouts-only allocations, the changes were -95.41% (95% CI -97.79% to -91.69%) and -95.10% (95% CI -97.73% to -90.10%).

A prospective local protocol then fixed the both-versus-neither contrast and evaluated 40 additional, disjoint networks. The primary change was -98.13% (95% CI -99.19% to -93.94%), with improvement in 37 of 40 networks. The secondary changes relative to stimulation-only and readouts-only allocations were -99.28% (95% CI -99.71% to -93.01%) and -95.48% (95% CI -97.58% to -90.42%). Every allocation records calibration activity; the stimulus response being predicted is withheld from fitting.

Whole-network activity error under other held-out inputs changed by +5.36% (95% CI -12.00% to +30.00%) in the first cohort and +8.05% (95% CI -23.95% to +39.06%) in validation. The clearer benefit for the specified stimulus response motivates examining which predictive uncertainty the endpoint observations constrain.

2.3 Endpoint information constrains response ambiguity

Forty additional networks, disjoint from both endpoint cohorts, tested whether endpoint observations narrow the responses permitted by comparably fitted parameter vectors. Every allocation used five recordings and a common five-start MAP estimator. For each readout we searched for the smallest and largest nonlinear response within a MAP-objective allowance of 2 above the best fit. These numerical feasible spans describe prediction ambiguity under a common data-and-prior constraint.

Relative to five nonendpoint recordings, joint endpoint recording changed the mean squared response-range width by -99.08% (95% CI -99.62% to -96.10%), with contraction in 40/40 networks. Its response-MSE change was -96.22% (95% CI -99.63% to -79.45%), with improvement in 38/40 networks. Raising the standard deviation of endpoint recording noise to four and sixteen times its original value changed the range metric by +471.43% (95% CI +178.12% to +999.62%) and +3384.93% (95% CI +1215.52% to +8268.23%), respectively, relative to ordinary-precision endpoint recording. The corresponding MSE changes were +156.17% (95% CI +24.32% to +928.87%) and +570.70% (95% CI +125.13% to +3169.11%). Noise levels were included correctly in the likelihood, and standardised noise draws were paired across conditions.

An information intervention distinguished endpoint-gain information from generic parameter knowledge. Keeping the five nonendpoint observations fixed, we revealed either the stimulation gain, all three readout gains, all four endpoint gains or four unrecorded nonendpoint gains, and refitted all remaining gains. Revealing both ends changed response MSE by -98.03% (95% CI -99.26% to -92.77%) relative to the four-ordinary-gain control, with improvement in 39/40 networks. The changes relative to knowing only the stimulation gain or only the readout gains were -92.19% (95% CI -98.98% to -79.73%) and -97.91% (95% CI -99.03% to -91.54%). These controls isolate the predictive value of endpoint-gain information.

Together, the feasible-response, precision and parameter-information comparisons support a specific mechanism in this model family: endpoint calibration constrains uncertainty relevant to the intended intervention response. This complements the earlier observation that accurate response prediction need not coincide with accurate recovery of each individual gain.

2.4 Endpoint gains and recurrent combinations have complementary predictive value

The finite stimulus response can be expressed as input, recurrent-propagation and readout factors (Methods). In the small-activity limit, the endpoint gains control how the stimulus enters and how the

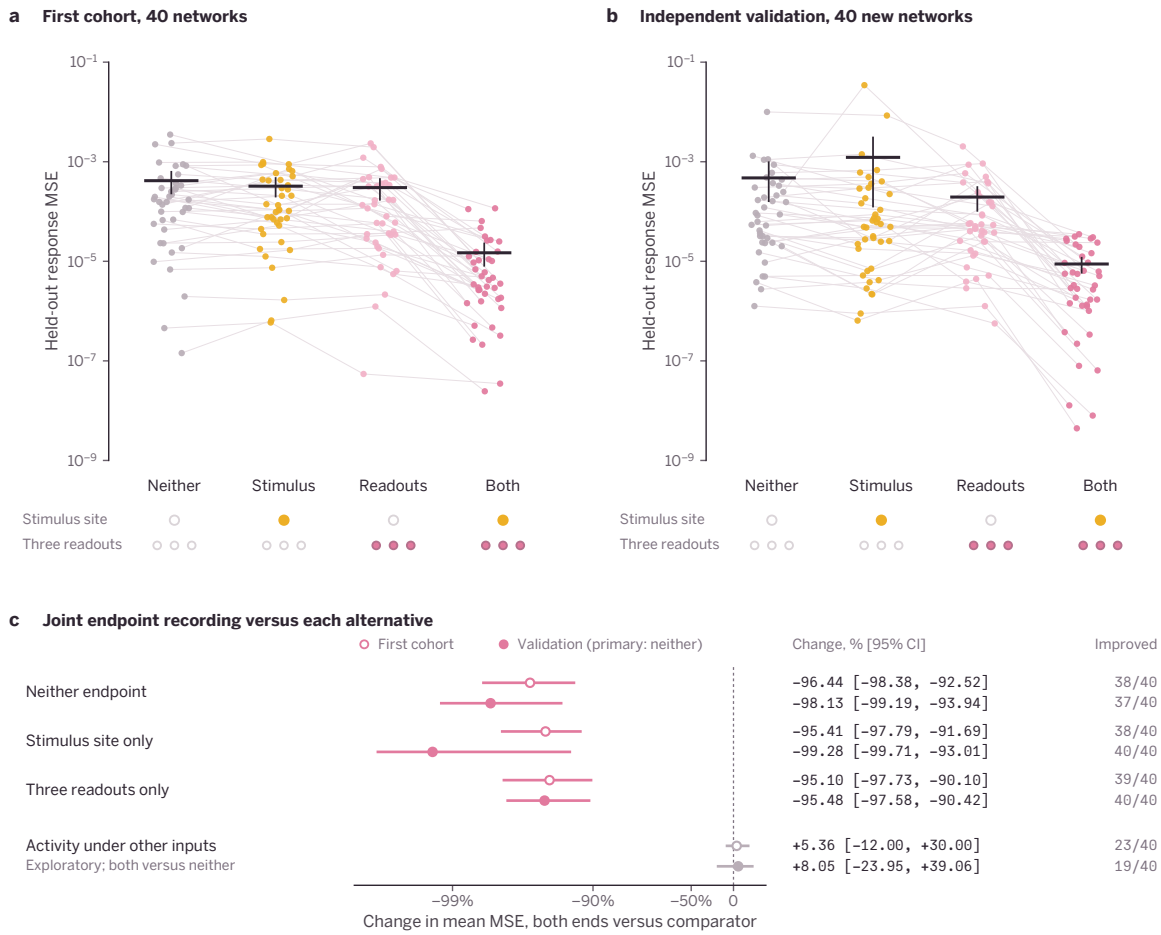


Fig. 1 | Endpoint allocation and held-out response prediction. a,b, Held-out response MSE for each of the first 40 networks (a) and 40 independent validation networks (b). All four allocations record five sites; the key below each panel marks whether the stimulation site and the three readouts occupy recording slots. Light lines join the same network across allocations; ink bars show arithmetic means and 95% network-bootstrap intervals. **c,** Change in aggregate mean MSE for joint endpoint recording relative to each alternative allocation, with paired 95% network-bootstrap intervals, in the first cohort (open markers) and the validation cohort (filled markers). The bottom rows compare both-end with neither-end recording for whole-network activity under other held-out inputs, an exploratory analysis. Both-versus-neither is the primary validation comparison; the other validation contrasts are secondary. Printed values give each change and interval; the right column counts networks with lower error.

response is read, while four connectivity-weighted sums of internal gains determine the two-dimensional recurrent dynamics. The endpoint dependence appears through stimulation/readout gain products. This identifies a concrete parameter-combination hypothesis behind the information result: calibration can constrain the response by constraining these factors.

Forty new networks tested the predictive value of this information with identical five-site ordinary calibration data and a common nine-start estimator. Revealing the four endpoint gains and four recurrent combinations changed response MSE by -98.34% (95% CI -99.87% to -93.56%) relative to revealing eight ordinary gains; all 40 networks improved. Relative to knowing the four endpoint gains alone, the additional recurrent information changed mean MSE by -91.37% (95% CI -92.52% to -32.80%), with improvement in 31/40 networks. The reduction remained between 74.81% and 91.71% when each network was omitted in turn. Thus internal propagation retains predictive importance even after endpoint gains are known. The second contrast supplies four additional constraints; the dimension-matched control below separates their specificity from the amount of information.

The exact factor decomposition was independently evaluated on fitted-versus-generating response differences in these same forty networks. Joint endpoint recording changed the mean squared input-factor contribution by -98.69% (95% CI -99.50% to -96.35%) and the readout-factor contribution by -98.79%

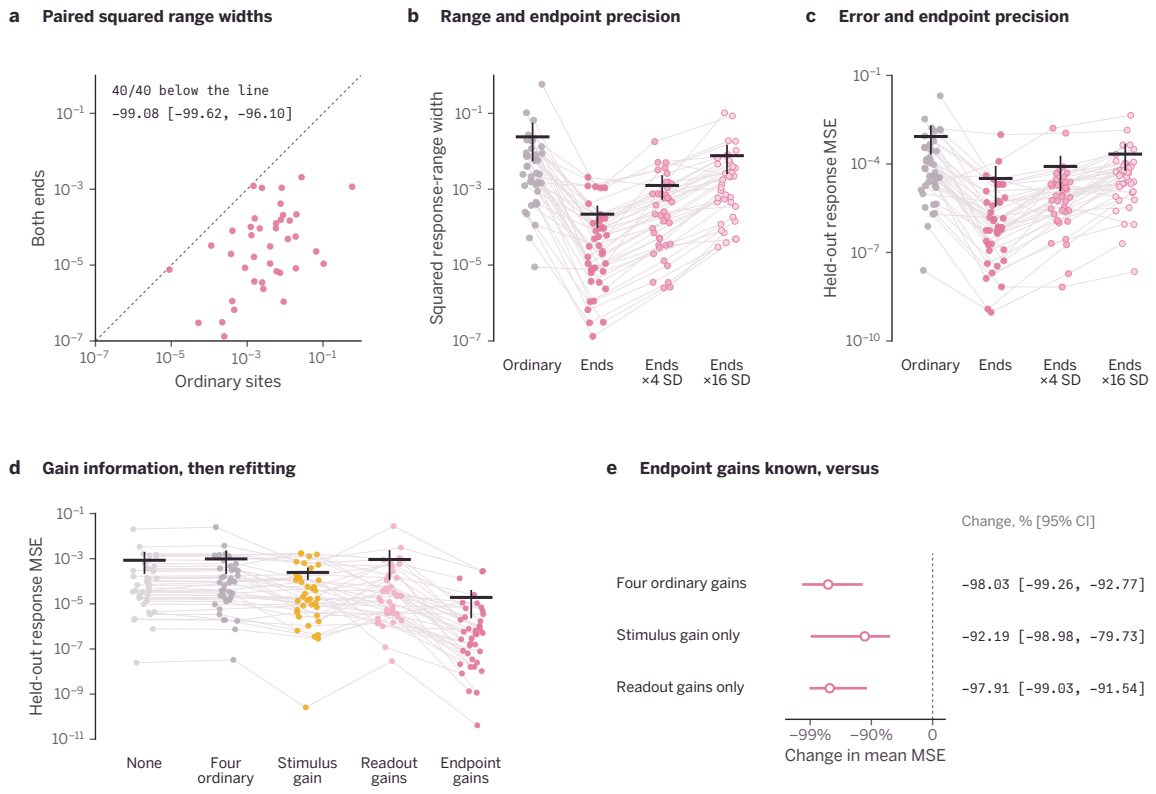


Fig. 2 | Endpoint information and response ambiguity. a, Mean squared numerical response-range width under a common MAP-objective allowance, with five ordinary recordings (horizontal) and joint endpoint recording (vertical), for each of 40 independent networks; the dashed line marks equality. This both-versus-neither comparison is primary. **b,c,** Range width (b) and held-out response MSE (c) for ordinary-site recording and for endpoint recording at the original noise level or with four or sixteen times its standard deviation (open markers). **d,** Held-out response MSE after revealing no gains, four ordinary gains, the stimulation gain, the three readout gains or all four endpoint gains and refitting the remaining gains, with the same five nonendpoint recordings in every condition. **e,** Changes in mean MSE when all four endpoint gains are known, relative to the alternatives in d, with paired 95% network-bootstrap intervals. In b–d, light lines join networks and ink bars show arithmetic means and 95% network-bootstrap intervals. Precision and oracle contrasts are secondary.

(95% CI -99.59% to -96.88%). The recurrent-factor change was +16.53% (95% CI -34.62% to +139.45%). This localizes the strongest error contraction to entry and readout, while retaining a role for internal propagation. The reconstructed response agreed with the original integration to a maximum absolute error of 2.20×10^{-15}.

A dimension-matched follow-up kept the same four endpoint gains known in both groups and compared four additional recurrent combinations with four additional ordinary gains. The recurrent combinations changed response MSE by -91.45% (95% CI -92.57% to -24.68%), with improvement in 32/40 networks. Leave-one-network reductions ranged from 72.73% to 91.80%. This comparison supports the predictive specificity of the internal combinations beyond the amount of revealed information. The follow-up was specified after the planned combination results, and uses the same forty networks.

The static combinations also captured much of the local response sensitivity. Their median squared-gradient projection fractions were 0.910 under ordinary recordings and 0.898 under endpoint recordings, compared with 0.289 and 0.287 for endpoint coordinates alone and 0.079 and 0.077 for eight random combinations. At parameter RMS perturbations of 0.1 prior SD, preserving the eight combinations reduced mean squared response changes relative to preserving endpoints alone by 42.12% (95% CI 28.97%–70.54%) and 28.97% (95% CI 5.24%–78.00%), respectively. At RMS 0.3, both corresponding intervals crossed zero (Supplementary Fig. 9). The static combinations provide an informative local approximation; finite nonlinear response changes also involve trajectory-dependent slopes.

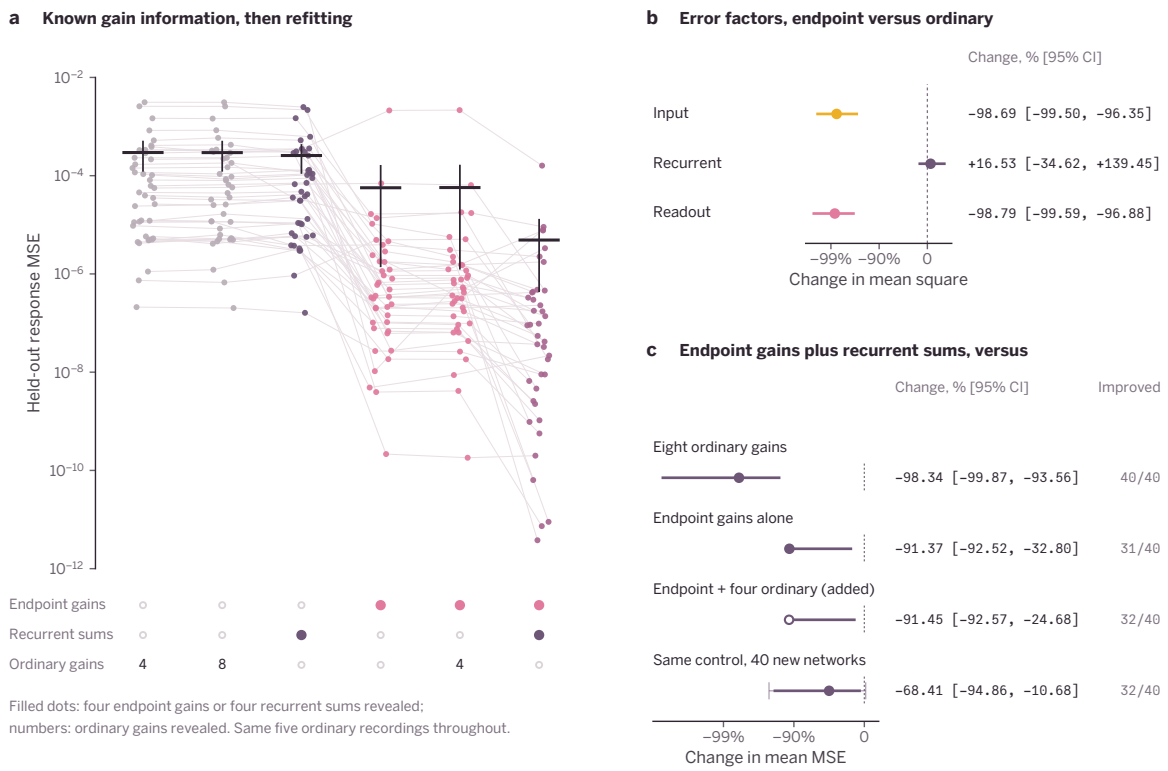


Fig. 3 | Parameter information and finite-response factors. a, Held-out response MSE after revealing specified gain information and refitting all free directions, with the same five ordinary recordings in every condition. The matrix below marks whether the four endpoint gains or the four recurrent gain sums were revealed, and how many ordinary gains. Light lines join the 40 networks; ink bars show arithmetic means and 95% network-bootstrap intervals. Results include additional common fitting for all six policies on one flagged network. **b,** Changes in the mean square of the input, recurrence and readout contributions to the fitted-versus-generating response difference, for endpoint relative to ordinary recording across the same forty networks. Contributions use all eight factor hybrids and the exact three-factor Shapley identity; hybrid factors give an algebraic decomposition of response differences. **c,** Change in mean response MSE when the endpoint gains and recurrent sums are revealed, relative to eight ordinary gains, the endpoint gains alone, and the endpoint gains plus four ordinary gains. The last control was added after the planned comparisons (open marker). The bottom row repeats it in 40 independent networks as a prespecified primary comparison; its thin interval is adjusted for that study's three primary comparisons. Intervals are paired 95% network-bootstrap intervals; the right column counts networks with lower MSE.

An independent forty-network test fixed this dimension-matched comparison before observing outcomes. Recurrent information changed response MSE by -68.41% (95% CI -94.86% to -10.68%), again improving 32/40 networks. Leave-one-network changes ranged from -91.96% to -57.84%. The interval adjusted across this study's three primary comparisons was -95.57% to +4.51% (Fig. 3c).

2.5 A fifth recording constrains recurrent-combination information

We next asked whether calibration recordings could acquire this information. On this independent cohort, the three five-site policies retained the four endpoints and selected one additional site using a random choice, an existing A-optimal criterion for the recurrent combinations, or the original response-risk criterion. Combination-targeted recording changed the squared feasible width along a direction fixed from the prior by -27.15% relative to a random fifth site (95% CI -40.09% to -15.18%; three-comparison-adjusted CI -43.18% to -12.69%). The range contracted in 30/40 networks. Selection used a local covariance approximation, whereas the subsequent range search used the nonlinear calibration model and a common MAP-objective allowance (Fig. 4b,d).

The corresponding response-MSE change was -8.95% (95% CI -41.49% to +33.68%; adjusted CI -48.91% to +43.09%), leaving the prediction benefit uncertain. Combination estimation MSE changed by -31.31% (95% CI -55.93% to +0.83%). In secondary comparisons with response-targeted recording,

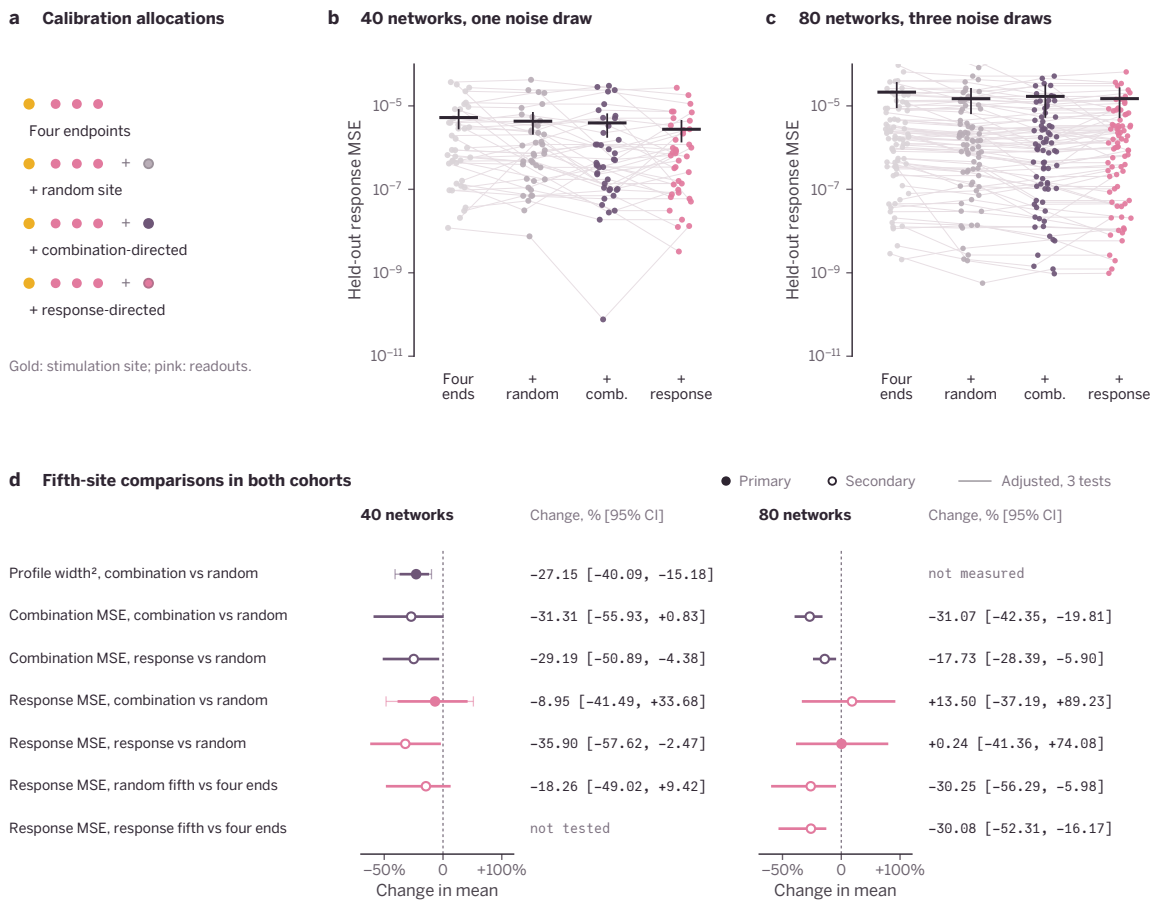


Fig. 4 | From parameter information to calibration recordings. a, Calibration allocations. The stimulation site (gold) and three readouts (pink) are recorded alone or with a fifth site chosen at random, by a local A-optimal criterion for the recurrent gain sums, or by the original response-risk criterion; the four-endpoint condition uses one fewer recording. **b,c,** Held-out response MSE for each network in the 40-network cohort with one noise draw (b) and the independent 80-network cohort with three noise draws (c). Light lines join networks; ink bars show arithmetic means and 95% network-bootstrap intervals. **d,** The same comparisons in both cohorts as changes in aggregate means, with paired 95% network-bootstrap intervals. Profile width^2 is the squared feasible width along one recurrent direction fixed from the public prior before calibration outcomes, under a common MAP-objective allowance; combination MSE uses four orthonormal internal coordinates. Filled markers are each study's prespecified primary comparisons; thin intervals are Bonferroni intervals for the three primary comparisons of the 40-network study, whose third primary comparison is the oracle contrast in Fig. 3c. Response-directed versus random selection, secondary in the 40-network cohort, is the single primary comparison of the 80-network cohort. Other comparisons are secondary.

combination targeting narrowed the squared profile width by 17.87% (95% CI 3.74%–31.22%) but gave a response-MSE change of +42.04% (95% CI -8.08% to +133.77%).

An independent follow-up evaluated eighty new networks with three noise repeats per network. In secondary comparisons, adding a response-directed or random fifth site reduced response MSE by 30.08% and 30.25%, respectively, relative to recording the four endpoints alone (Fig. 4c,d; Supplementary Table 3). Both allocations improved the noise-averaged error in 60/80 networks. The prespecified primary comparison, response-directed versus random fifth-site selection, gave +0.24% mean MSE (95% CI -41.36% to +74.08%).

The same follow-up strengthened the evidence for acquiring internal information through recordings. Combination-directed selection reduced actual estimation MSE in the four internal coordinates by 31.07% relative to a random fifth site (95% CI -42.35% to -19.81%), with improvement in 63/80 networks (Fig. 4d). The corresponding comparison with response-directed selection also favoured combination estimation. These secondary results extend the earlier feasible-range finding to more accurate estimates of the internal combinations. Supplementary Table 3 gives the complete comparisons; the accompanying analysis examines how internal precision relates to response error.

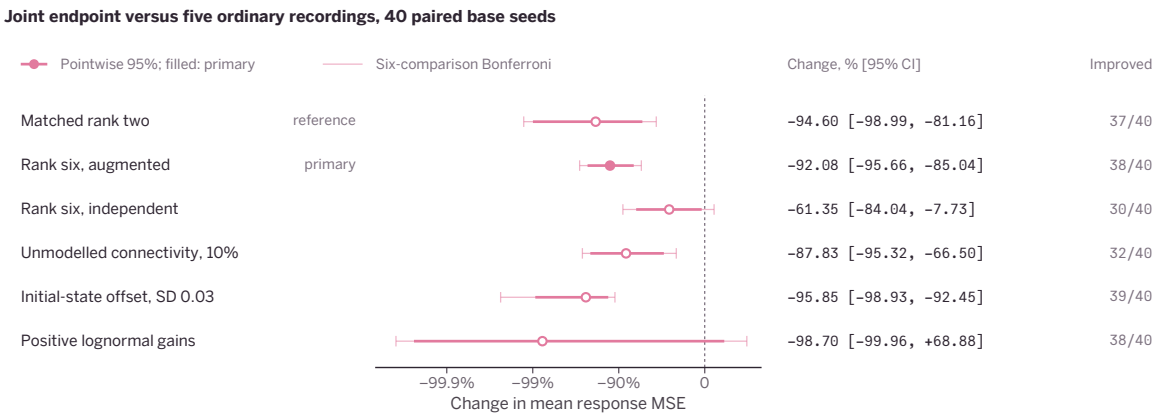


Fig. 5 | Structural and model-assumption sensitivity. Changes in aggregate mean response MSE for joint endpoint recording relative to five ordinary recordings. Each condition contains forty independent base seeds; the six conditions remain paired within seed. Thick intervals are pointwise 95% network-bootstrap intervals and thin intervals are six-comparison Bonferroni intervals. Rank-six augmentation is the planned primary comparison (filled marker). The right column counts base seeds with lower MSE.

2.6 Endpoint allocation retains value under structural and model perturbations

The same forty base seeds supplied six paired model conditions, each comparing five ordinary recordings with joint endpoint coverage. In the prespecified rank-six augmentation, endpoint recording changed response MSE by -92.08% (95% CI -95.66% to -85.04%), with improvement in 38/40 networks. Adding an unmodelled connectivity term at 10% relative Frobenius norm gave a change of -87.83% (95% CI -95.32% to -66.50%), and an unknown initial-state offset of SD 0.03 gave -95.85% (95% CI -98.93% to -92.45%). These effects, and the matched rank-two reference (-94.60%), retained negative six-comparison Bonferroni intervals (Fig. 5).

The independently generated Gaussian rank-six condition gave -61.35% (95% CI -84.04% to -7.73%; 30/40 networks improved), with a multiplicity-adjusted interval spanning zero. Positive lognormal gains gave a large point estimate (-98.70%; 38/40 networks improved) but a wide 95% interval (-99.96% to +68.88%). The latter mean comparison is sensitive to the heavy-tailed errors, despite an improvement in most networks. These conditions extend the evaluation beyond the original signed-gain rank-two family while showing that effect magnitude and certainty depend on model structure and parameterization.

2.7 Endpoint coverage provides a baseline for recording criteria

The simple endpoint baseline also changed the interpretation of the risk-based selectors. In the first 40 networks, nominal selection versus this baseline gave +23.64% (95% CI -41.31% to +161.94%), and ensemble selection gave +260.80% (95% CI -56.39% to +1250.22%). These intervals leave their incremental benefit over endpoint recording uncertain.

The original prespecified nominal-versus-ensemble comparisons did not establish an additional error reduction. In the independent 100-unit confirmation, the primary shared-response contrast gave +7.95% mean MSE (95% CI -23.59% to +20.95%). The original 12-unit error comparisons and separate risk-forecast test also retained intervals spanning zero; their complete results, including the initially favorable pilot, are reported in Supplementary Figs. 10–14 and Supplementary Table 1. Selection overlap and matched-fitting diagnostics help interpret these comparisons (Supplementary Figs. 6 and 12).

When endpoint access or budget was restricted, the covariance-based comparison remained uncertain after the matched-fitting check (Supplementary Fig. 7). Endpoint-pair lookahead repeated ensemble-greedy choices in 237/240 budget/precision settings and added negligible predictive benefit (Supplementary Fig. 8). These comparisons make endpoint coverage an informative baseline for assessing what further selection procedures contribute.

3 Discussion

Joint calibration of the stimulation and readout sites improved prediction of the held-out stimulus response across independent networks. The benefit remained under rank-six augmentation and mild errors in connectivity or initial state. The benefit was clearest for the specified stimulus-to-readout response, supporting the choice of calibration recordings according to the response to be predicted.

The response-range experiments clarify what changed. Models with similar calibration fits could disagree widely about the stimulus response when endpoints were unobserved. Endpoint recordings narrowed that disagreement; reducing their precision increased both the response range and prediction error. Revealing endpoint gains and refitting the remaining parameters supplied a separate test of the information those sites carry. Together, these comparisons connect endpoint calibration to narrower uncertainty in the intended response.

The factor decomposition locates the strongest contraction in the input and readout contributions to response error. In the low-activity limit, endpoint gain products and four recurrent gain combinations determine the transfer response. The information controls support additional predictive value for those recurrent combinations after endpoint gains are known, with the same direction of improvement in two independent cohorts of the equal-dimensional comparison. Earlier connectome-constrained models distinguish stiff parameter combinations from poorly recovered individual gains [9], as does the broader literature on sloppy models [8]. Our experiments identify and test the conditional value of particular combinations for this held-out response. Their static form captures much of the local sensitivity; larger nonlinear changes also depend on activation slopes along the trajectories.

The fifth-recording studies connect the parameter-information controls to observations. With the four endpoints already recorded, combination-directed selection narrowed a nonlinear feasible range. The independent follow-up then showed more accurate recovery of the internal combinations relative to a random fifth site. Additional calibration also improved response prediction beyond four-endpoint coverage. Together, these experiments connect the parameter combinations identified by the model to information that calibration recordings can acquire.

The endpoint allocation result also gives the method comparisons a concrete baseline. Recording-site optimisation and goal-oriented design already value observations according to a prediction task [9, 20, 22, 26], and prediction profiles provide tools for examining uncertainty in that prediction [27–30]. Active photostimulation design chooses informative inputs and has been evaluated on real neural recordings [31]. For our fixed-input task, endpoint coverage explains a large gain over random recording. Comparing selectors against this baseline helps distinguish the benefit of endpoint access from the added value of the selection procedure.

3.1 Model scope

The experiments concern simulated networks with known connectivity and initial states, a specified stimulus and rate observations with Gaussian noise. Structural follow-ups examine rank-six networks and mild connectivity or initial-state errors. Endpoint allocation requires access to the stimulation and readout sites during calibration. The finite-response factorization is exact within the model, while the static gain combinations describe its small-activity limit. The numerical feasible ranges quantify responses or parameter values compatible with a specified MAP-objective allowance. These assumptions define the setting for the prediction and information results.

3.2 Implications for recording-design practice

An informative baseline should reflect the known stimulus and readouts. When endpoints are available during calibration, comparing a risk-based selector only with unrestricted random sampling can substantially overstate the value of sophisticated selection. The endpoint baseline offers a direct, inexpensive comparison. Endpoint accessibility should therefore be stated explicitly when comparing recording policies.

4 Conclusions

Joint calibration of stimulation and readout sites improves prediction of a held-out stimulus response in the tested networks, with replication across independent cohorts and support under several structural and model perturbations. Endpoint information narrows the range of compatible responses and reduces input and readout error contributions. Information controls support additional predictive value in recurrent gain combinations. A targeted fifth recording constrains a prespecified internal direction and improves estimation of the recurrent combinations on independent networks. Adding a fifth recording improves prediction relative to four-endpoint calibration. Endpoint coverage provides a useful baseline, and the recurrent combinations provide a concrete target for additional calibration.

5 Methods

5.1 Notation

Symbol	Meaning
J, N	known connectivity matrix and number of units
$\theta = (\eta, \mathrm{b})$	uncertain log-gains and biases in the 12-unit model
g, τ	neuronal gains and time constant in the 100-unit model
L, R	low-rank connectivity factors in J = LR
M	recording budget (number of recorded neurons)
q, Q	specified target response and the three-response vector
F, H	whitened observation sensitivities and local information matrix
R(M)	local delta-method risk of a recording set
v, e	forecast of the mean squared error and the realised prediction error
S(v, e)	forecast score $\log \mathrm{v} + \mathrm{e}^2/\mathrm{v}$
LS, ES, LV, EV	nominal and ensemble criteria for the single-response and three-response objectives

5.2 Main 100-unit model and calibration task

The endpoint, mechanism and fifth-recording studies use the same reference model [9],

$$\tau \dot{x} = -x + Jr + u(t), \qquad r_i = g_i \tanh(x_i), \qquad J = LR. \tag{1}$$

Time and activity are dimensionless. The network has 100 units, time constant $\tau = 0.5$, zero biases and independent signed gains $g_i \sim N(1, 0.9^2)$. Connectivity, initial states and calibration inputs are supplied to the fitted model; the gains are inferred from noisy rate observations. The main response prediction concerns a separate stimulation trial that is withheld from calibration.

For each network, independent vectors $v_i \sim N(0, \Sigma)$ generate the rank-two factors $L_{ia} = v_{ia}/N$ and $R_{ai} = v_{i,a+2}$ for $a = 1, 2$. The covariance is

$$\Sigma = \begin{pmatrix} 1 & 0 & 1.5 & 3 \\ 0 & 1 & -3 & 1.5 \\ 1.5 & -3 & c & 0 \\ 3 & 1.5 & 0 & c \end{pmatrix}, \qquad c = 1.1^{27}. \tag{2}$$

This implements the covariance-inflation recipe used for the published model family [9]. The factor $1/N$ supplies the connectivity scaling; the resulting matrix retains its diagonal and is used without a further norm rescaling. Each initial-state entry is independently drawn from $N(0, 0.1^2)$ and the same known initial state starts both calibration trials and both target-response trials.

Three distinct calibration channels c_j are drawn from the network. The two trials, indexed by $k = 0, 1$, apply $u_{c_j}^{(k)}(t) = 0.4 \sin[(0.6 + 0.3j)t + 0.7k]$ for $j = 0, 1, 2$, with zero input on other channels. These inputs are evaluated on the 0.05 time grid and held constant between grid points. Calibration channels may overlap the task endpoints. The stimulation site s and three distinct readouts j_1, j_2, j_3 are selected using structural reachability in J, before generating gains or observation noise; each readout differs from s.

The held-out trial applies $u_s(t) = 0.6$ during $1 \le t < 2$, with all other inputs zero. Its matched control has zero input throughout. The three targets are

$$q_k(g) = \int_0^6 [r_{j_k}(t; g, u_*) - r_{j_k}(t; g, 0)]\, dt. \tag{3}$$

Both design and evaluation weight these three raw responses equally. The common reference settings are collected below.

Setting	100-unit reference configuration
Calibration duration	Two trials, each from time 0 to 6
Observation times	0, 0.2, ..., 5.8; 30 rate samples per trial and 60 per recorded neuron
Observation noise	Independent additive Gaussian noise, standard deviation 0.05 in rate units
Target response	Pulse amplitude 0.6 on one site during time 1-2; response difference integrated to time 6
Initial state	Known, independent entries from $N(0, 0.1^2)$; shared across trials within a network
Numerical integration	Input grid 0.05; two RK4 substeps of 0.025 per grid interval
Response quadrature	Rate differences and complete sensitivities integrated at the RK4 stages
Main recording allocation	Five sites: stimulation site, three readouts and one ordinary site; compared with five ordinary sites
Prior-averaged selection	32 shared prior draws; gains inferred in coordinates $z = (g - 1)/0.9$

Integration uses the exact state reduction $x = v + Lz_{\text{state}}$, analytic filtering of the held known input in v, and RK4 for the reduced state and gain sensitivities. Sampling therefore stays on the stated observation grid while integration and response quadrature use the finer steps.

These model, input, observation and integration settings are shared across the reference 100-unit cohorts. The follow-up subsections specify changes in recording budget, gain-prior width, noise level, connectivity, initial-state knowledge and gain parameterization. They also give the cohort-specific repetition and fitting rules. The executable definitions are `code/models.py` (`public_case`) and `code/rank2_refined.py` (`refined_case`) within `work/recording_extensions`; the shared scalar settings are in `protocols/confirmation.json`, with later study protocols in the corresponding round directories. The original 12-unit softplus model and its development and forecast protocols are described with the original criterion-comparison studies in the Supplementary analyses.

5.3 A factorial comparison of existing criteria

In whitened parameter coordinates, the local information matrix for a recording set M at parameter particle ℓ is

$$H_{M,\ell} = I + \sum_{i \in M} F_{i,\ell}^T F_{i,\ell}, \tag{4}$$

where F includes the observation derivatives, prior parameter scale and noise standardisation. With $a_{k,\ell}$ denoting the whitened target gradient, the scalar criterion is

$$R_{k,\ell}(M) = a_{k,\ell}^T H_{M,\ell}^{-1} a_{k,\ell}. \tag{5}$$

The vector criterion averages these risks over the three readouts. Local designs evaluate at μ=0; ensemble designs average the individually solved risks over 16 common prior draws. Averaging information before solving would define a different criterion and is not used.

Design	Predictive objective	Parameter representation
LS	One specified response	Nominal parameter
ES	One specified response	Common prior ensemble
LV	Three-response vector	Nominal parameter
EV	Three-response vector	Common prior ensemble
Activity	Full calibration activity	Nominal parameter
Random	No risk objective	Five feasible random sets

All risk-based methods use the same greedy forward selection and deterministic tie convention. These are greedy-selected sets, not certified global optima. Random permutations give nested budgets. The design API receives only the public model, prior, inputs and task. It receives neither generating parameters nor measured evaluation outcomes. There is no uncounted all-neuron pretraining or preliminary dataset.

The ensemble criterion is a **prior average of local delta-method risk**. It does not integrate future measurements and nonlinear posteriors, establish a variance upper bound, or capture all posterior modes.

5.4 Common inference and paired evaluation

After selection, the 100-unit studies fit gains by minimizing the Gaussian MAP objective

$$\mathcal{L}(z) = \frac{1}{2}\sum_{k,t}\sum_{i\in M}\left[\frac{r_i(t;1+0.9z,u^{(k)}) - y_{kti}}{\sigma_i}\right]^2 + \frac{1}{2}\|z\|^2. \tag{6}$$

The sum includes the observed times in both calibration trials. Analytic sensitivities supply the least-squares Jacobian. The parameter-combination, structural-robustness and fifth-recording studies use nine common starts and retain the smallest objective, with a 400-evaluation cap per start; earlier fitting rules and their sensitivity checks are specified below. The original 12-unit study uses its corresponding log-gain/bias prior, a prior-mean start and an 80-evaluation cap. Finite unconverged results are retained with flags, and exceptions or nonfinite predictions use a flagged prior-mean fallback.

Within each network, true parameter vector and noise repetition, methods receive their selected subsets of one common measurement-noise field. The full field is internal to the generator. Identical recorded datasets receive the same MAP objective, starting points, noise field and solver settings, and reuse the same fitted solution. Responses are predicted by nonlinear simulation at the fitted parameter. For the 100-unit activity-prediction diagnostic, two held-out trials use the same calibration channels with inputs $0.4\sin[(0.9+0.2j)t + 0.4k + 1.3]$, where $j = 0, 1, 2$ and $k = 0, 1$; these observations are withheld from fitting.

5.5 Estimands and uncertainty

For each network, squared response errors are averaged across the prescribed physiological draws, noise repeats, targets and random-design repetitions. Networks then receive equal weight in the aggregate MSE. RMSE is the square root of that aggregate. The reported percentage change is the ratio of aggregate mean MSEs minus one, rather than the average of individual percentage changes.

Uncertainty is estimated by resampling whole paired networks, retaining their nested repeats. The follow-up protocols identify the primary contrasts, bootstrap counts and any multiplicity adjustment. The structural and original fifth-recording studies report both pointwise and multiplicity-adjusted intervals; the independent fifth-recording follow-up has one primary contrast. Other intervals are pointwise 95% unless specified. Leave-one-network summaries assess the influence of individual networks. An interval spanning zero describes uncertainty about the direction of the mean change [32, 33]. The earlier 12-unit comparisons and their forecast-specific estimands are detailed in the Supplementary analyses.

5.6 Numerical and published-model checks

The implementation is checked against an independent NumPy model, analytic finite differences, nonordered observation subsets, a linear bias-cancellation null, and exact small-subset enumeration. A separate nondegenerate linear-Gaussian positive control uses constant observation/target Jacobians, independent prior truths and noise, and exact linear MAP predictions. It checks both nominal–ensemble equivalence and the link between $a^{\mathrm{T}}Ca$ and actual MSE. The original SVD criterion is independently

calibrated in its noiseless linear internal-state setting against Monte Carlo reconstruction, rather than mislabelled as a noisy-rate baseline reproduction.

The N=100 model-family extension uses the published rank-two tanh distribution, raw signed gains g~N(1,0.9^2), zero bias and τ=0.5 [9]. Its inputs, known initial state, measurement noise and inference are explicit adaptations. Negative gains are retained as mathematical parameters. The coarse Euler implementation did not pass continuous-time convergence checks. The refined implementation uses x=v+Lz, analytic filtering of the known held input, and RK4 on the reduced state and sensitivities. Responses and their complete gradients are integrated at RK4 stages. It is compared with dense RK4 and finer substeps before execution. The initial pilot used eight new networks, two truths, two noise repetitions, three targets, budgets 5/10 and both pools. The prior ensemble has 32 draws and MAP permits 200 evaluations. These settings are frozen before formal extension outcomes, following numerical and timing checks. All initial pilot comparisons are exploratory. This extension changes both model family and N, so it is not a controlled estimate of an N-only scaling law.

5.7 Independent confirmation and focused follow-ups

The 100-unit confirmation used 40 new networks (IDs 20000–20039), two independent physiological draws, two noise draws and three fixed readouts per network, both recording pools and budgets five and ten. The primary contrast was ensemble versus nominal selection for the shared objective, with endpoints excluded and budget five. The original model, integration, noise, 32-particle design ensemble and common MAP rule were retained. The sample size and primary contrast were locally frozen before inspecting outcomes. All other confirmation contrasts use exploratory 95% network-paired bootstrap intervals.

The matched-width analysis reused these networks with prior and truth standard deviations 0.3, 0.6 and 0.9, scaling common normal draws. First-step rank correlations used a common empty selection prefix. Component diagnostics held either target sensitivities or observation information at the nominal values. The fitting follow-up kept the selected sets fixed and fitted from zero, two draws scaled to 0.25 prior standard deviations and two full prior draws. Starts were common to both selectors and data repetitions within a network (seed stream 91631). The fit with the smallest unchanged MAP objective supplied the response prediction. True parameters were used only for diagnostics: a returned objective above the objective at the generating parameters certifies a suboptimal returned fit, but the reverse does not certify optimality.

The endpoint experiment fixed five recordings and used four allocations: five nonendpoints; the stimulus site plus four nonendpoints; the three readouts plus two nonendpoints; and all four endpoints plus one nonendpoint. For each of three allocation repeats, a common random permutation of nonendpoints supplied the remaining sites (streams 1601–1603). Thus policies had identical total measurement counts and shared truth and observation-noise draws. The original zero-start MAP estimator was used throughout. The first cohort reused networks 20000–20039. A separate prospective local protocol fixed both-versus-neither as the primary comparison on 40 new networks (IDs 30000–30039); both-versus-stimulus-only and both-versus-readouts-only were secondary comparisons. All 40 planned networks were retained in each cohort.

Response squared errors were averaged over targets, truth draws, noise draws and allocation repeats within each network. Contrasts are ratios of means across networks minus one, with 10,000 paired network bootstrap resamples. Evaluation rows and fit restarts are not treated as independent observations. Local freezes and code hashes accompany the outputs; they are not public preregistrations.

For the local information diagnostic, parameters were prior-whitened and the nominal Gaussian covariance was C=(I+F_S^{T}F_S)$^{-1}$. Marginal variances were compared with the unit prior variance. If an endpoint parameter block E were known exactly, the conditional covariance would be C−C_:E C_EE^{-1} C_E:. The reduction in trace(GCG^{T})/3 quantifies the local risk removable by that additional knowledge, including parameter correlations. This is an approximation diagnostic rather than an empirical variance estimate. A separate nonlinear oracle check substituted true endpoint gains into fitted parameter vectors, held all other fitted values fixed and recomputed the response; oracle outcomes never entered policy or fit selection.

Task specificity was evaluated by simulating the already fitted models under the original other-input test traces and comparing whole-network activity with the teacher. For the descriptive response-error decomposition, write q_i=$g_i$$L_i$, where L_i is the time-integrated difference of tanh states under stimulated and control trajectories. Then $\hat{q}_i - q_i = (\hat{g}_i - g_i)L_i + \hat{g}_i(\hat{L}_i - L_i)$. We report the mean square of each

term and twice their mean product. The cancellation fraction is minus this cross term divided by the sum of the two squared terms, aggregated over the same records as the response MSE. The sum agreed with the response MSE for every saved fit; the decomposition describes compensation in fitted models rather than a separate intervention on biological dynamics.

5.8 Response ambiguity and gain-information interventions

The mechanism study used network IDs 40000–40039, one prespecified physiological draw and noise draw per network, and the same three readouts. Five common starts comprised the nominal mean, two draws at one quarter of the prior standard deviation and two full prior draws (stream 93631). Fitting retained the smallest MAP objective with a 300-evaluation cap. The response-range problem maximised or minimised each target over the full nonlinear MAP sublevel set with allowance delta=2. SLSQP used two starts per direction, a 160-iteration cap, analytic sensitivities and invertible local Gauss-Newton coordinates. The quadratic prior makes the original finite coordinate bounds redundant. Every retained extreme was checked against the original objective; the final numerical envelope also included every known feasible multistart fit, preserving the original search output separately; discovering a lower objective triggered refitting and a repeated range search. Value-only line searches and the coordinate transformation preserved the scientific objective, feasible region, samples and comparisons; their numerical checks and timing amendments were logged before completed validation outputs.

The primary range metric averaged squared marginal response widths across the three targets, then across networks. Secondary experiments multiplied endpoint-noise standard deviations by four and sixteen while keeping their standardised random draws paired. The first eight IDs were fixed for a delta=5 threshold check. The joint-endpoint versus ordinary-site range-metric change was -99.77% (95% CI -99.85% to -93.61%) in that check. Known feasible delta=2 extrema were retained in the wider search envelope. All retained extrema were reevaluated with twice the integration resolution; the maximum response change was 0.00231 of the corresponding range width. The primary comparison retained 479 successfully optimized boundary points and one MAP point. Two of 480 retained points in the noise conditions and two of 96 points in the wider-threshold check had unsuccessful optimizer flags but satisfied the numerical feasibility tolerance. The finer integration changed the primary range-metric effect from -99.08477% to -99.08468%. Ranges are feasible numerical spans, not guarantees of global extrema or calibrated confidence intervals.

Oracle information controls used the same five nonendpoint recordings and fixed the revealed gains to their generating values before refitting every remaining gain. The matched ordinary-gain control revealed four gains outside both the endpoints and the recorded set (stream 93931). These diagnostic gains never entered a recording selector. Statistical comparisons resampled complete networks 10,000 times, keeping all within-network contrasts paired.

5.9 Indirect-recording comparison using joint prior simulations

A separate locally frozen comparison used networks 45000–45039, two physiological draws, two noise draws and three targets per network. All selectors chose five nonendpoint sites and used the same five-start nonlinear MAP estimator. In addition to nominal and prior-averaged local-risk selection, a joint covariance criterion estimated observation–response dependence from 32 prior simulations. A held-out-prior variant evaluated the resulting linear predictor on 32 independent prior simulations and added the analytic contribution of observation noise to its squared error. Greedy choices minimised this surrogate risk. No realised calibration or held-out stimulus response was available during selection. This construction draws on established goal-oriented and simulation-based design ideas [20, 25, 34].

The held-out-prior versus nominal MSE contrast was primary. Covariance, ensemble and random comparisons were secondary. A retrospective optimization audit identified 11 truth/noise datasets in which at least one selected fit had an objective above that at the generating parameters. All five policies for each flagged dataset received the same eight additional prior-generated starts (stream 94031), plus refinement of the old best fit. The generating parameters flagged the diagnostic but were never starting points or selection criteria. These sensitivity results are reported separately from the frozen comparisons. The final response estimates always came from the common nonlinear fit, so the comparison evaluates whether the surrogate selects useful recordings for that estimator. Numerical tests checked the dual covariance formula against direct observation-space calculations and verified the response and objective derivatives.

5.10 Finite-response factors and parameter-combination controls

For the rank-two family, write the known connectivity as $J = LR$ and each trajectory as $x^a = v^a + Lz^a$, where $a = 0, 1$ denotes control or stimulation and $\tau \dot{v}^a = -v^a + u^a$ includes the known initial state. Define the diagonal secant matrix $D(t)$ by

$$D_{ii}(t) = g_i \frac{\tanh x_i^1(t) - \tanh x_i^0(t)}{x_i^1(t) - x_i^0(t)}, \tag{7}$$

with the continuous derivative limit when the denominator vanishes. Because only site s is stimulated, the exact finite-response equations are

$$\tau \dot{\delta z} = (-I + A(t))\delta z + b(t), \qquad q = \int_0^T C(t)\delta z(t)\, dt, \tag{8}$$

where $A = RDL$, $b = R_{:s} D_{ss} \delta v_s$, and $C = D_{\mathcal{R}\mathcal{R}} L_{\mathcal{R}:}$ for the three readouts $\mathcal{R}$. The input factor, recurrent propagation and readout factor therefore depend on distinct combinations of gains and trajectory-dependent activation slopes. This follows the established low-rank dynamical representation [35]; the secant construction retains the finite stimulus amplitude. Coefficients were saved at the same RK4 stages as the original trajectories. Reintegrating these equations recovers the original response without a small-stimulus approximation.

To locate response disagreement, each retained profile extreme was compared with its corresponding MAP reference in the earlier forty-network mechanism cohort. All eight combinations of reference/extreme input, propagation and readout factors were integrated, and three-factor Shapley contributions assigned their combined response difference. These hybrid paths are a functional decomposition, not physical neural interventions. Squared contributions are not additive fractions of total prediction MSE because cross terms remain. A new forty-network cohort supplied a separate decomposition of fitted-versus-generating response errors.

In the small-activity limit, D approaches $\mathrm{diag}(g)$, suggesting four endpoint gains and four entries of $A_0 = R\,\mathrm{diag}(g)L$ as candidate predictive combinations. In that limit, the stimulus-to-readout transfer has the form

$$\widehat{\delta r_o}(p) = g_o L_{o:} \left[(\tau p + 1)I - A_0\right]^{-1} R_{:s} g_s \frac{\widehat{u}_s(p)}{\tau p + 1}. \tag{9}$$

Here p is the Laplace variable. Thus the endpoint gains set the input and output factors, while the four weighted sums $(A_0)_{ab} = \sum_i R_{ai} L_{ib} g_i$ determine the intervening two-dimensional dynamics. Endpoint dependence enters through the products $g_s g_o$, making the joint input/readout uncertainty explicit. This explains the candidate combinations without requiring recovery of every internal gain; the eight revealed quantities are not asserted to be a minimal parameterization. At finite activity, the exact factors above also depend on trajectory-specific slopes. We tested the static approximation explicitly. In prior-standardized parameter coordinates, the response Jacobian was projected onto the row space of these eight linear combinations, onto four endpoint coordinates alone, or onto eight fixed random linear combinations. Finite perturbations used eight common random directions and both signs, projected into each constraint null space and normalized to parameter RMS changes of 0.1 or 0.3 prior standard deviations. These interventions preserve the specified combinations exactly but do not impose a calibration-likelihood constraint.

Independent gain-information controls used network IDs 50000–50039 and retained five ordinary calibration recordings for every policy. Before refitting, they revealed the four endpoint gains, the four recurrent combinations, both groups, four ordinary gains, or eight ordinary gains. Ordinary controls excluded both endpoints and recorded sites. A singular-value decomposition parameterized the exact constraint null space; the same nine starts were projected into each space. The two specified comparisons were joint eight-combination information versus eight ordinary gains, and its additional value over four endpoint gains. The latter comparison measures the value of extra information and does not hold its dimension fixed. These networks also form the robustness cohort below and are counted once. A retrospective dimension-matched follow-up revealed the same four endpoint gains plus the same four ordinary gains used in the original ordinary-gain control. All forty networks were retained. Any selected fit with an unsuccessful termination or objective above the known feasible generating-parameter objective triggered a matched fitting audit. On the one flagged network, all six oracle policies received seventeen additional common starts: the nine fits to the same ordinary calibration data without oracle knowledge

and eight new fixed prior draws, projected into each constraint space. The original nine fits remained candidates. This removed the feasible-cost failure; all final selected oracle fits terminated successfully and satisfied that check. The dimension-matched effect changed from -91.46% to -91.45%. Unrevealed true gains were not supplied as starting values.

5.11 Network structure and model-assumption perturbations

Forty independent base seeds (50000–50039) were evaluated in six paired conditions at budget five. The matched rank-two model supplies a reference. A rank-six augmentation adds four independent Gaussian modes at one half the original connectivity Frobenius norm, then rescales the combined matrix to the original norm. A separately generated Gaussian rank-six matrix also matches that norm. The public stimulation site, three readouts, calibration inputs, standardized gain draws and noise draws remain paired across these structural conditions.

Two separate misspecification conditions change only the generating model: an unmodelled rank-four connectivity term of relative Frobenius norm 0.1, or an unknown initial-state offset with independent entries of SD 0.03. The fitted model retains the original connectivity or initial state, respectively. A final matched condition replaces signed raw gains with positive gains $g = \exp(\theta - 0.9^2/2)$, retaining a Gaussian latent prior $\theta \sim N(0, 0.9^2 I)$ and applying the exact chain rule in fitting.

Each condition compares joint endpoint recording with five ordinary recordings, with one physiological draw, one noise draw and three response targets per base seed. All fits use nine common starts (the prior mean, two quarter-scale and six full prior draws), a 400-evaluation cap, and the same MAP objective. The rank-six augmentation contrast is primary; remaining conditions are secondary. Paired bootstrap summaries include 95% intervals, six-comparison Bonferroni intervals and leave-one-seed influence. Paired structural variants are not counted as additional independent sampling units. Reevaluation with twice the integration resolution changed the six MSE effects by at most 0.00068 percentage points and the three main combination-information effects by at most 0.00042 percentage points. Independent derivative and integration checks preceded the experiments. All selected fits terminated successfully and achieved objective values below their known feasible reference.

5.12 Limited-budget allocation and endpoint-pair lookahead

A separate forty-network cohort (55000–55039) tested budgets two and three under uniform recording noise, fourfold stimulus-site noise SD, and fourfold readout-site noise SD. All noise levels are supplied to both selection and fitting. Six policies were evaluated using the same calibration draws and nine-start MAP rule. The candidate augments ordinary ensemble-risk greedy selection with each of the six endpoint-pair seeds. At budget three, each pair receives its best additional site from the complete candidate pool. The final set minimizes the existing ensemble local-risk objective over these candidates and the original greedy set.

Comparators are ordinary ensemble and nominal greedy selection over all sites; exact enumeration of all endpoint-only subsets; a stimulus-first policy filling remaining slots with a fixed random ordering of readouts; and random all-site recording. The ensemble objective and its 32 prior particles are unchanged. Risk evaluation solves only for the three response-gradient right-hand sides, which is algebraically equivalent to the original full-inverse computation. Identical recording sets reuse the same fit. Shared prior-bank construction and subsequent selection timings are reported separately.

The primary comparison is the candidate versus ordinary ensemble greedy selection, with response MSE averaged equally over the six budget/precision strata within each network. Performance against exact endpoint-only enumeration is an additional required comparator. Individual strata and other comparisons are secondary. Enumeration, batch selection and improvements to greedy design are established approaches [36]; this experiment tests whether endpoint-informed search improves actual nonlinear predictions in this task.

5.13 Acquiring recurrent-combination information through recordings

A further cohort of forty networks (60000–60039) tested the dimension-matched information contrast and the acquisition of recurrent information through calibration recordings. The protocol and numerical sources were fixed before generating this cohort's outcomes. The information comparison retained five

ordinary recordings and revealed either four endpoint gains plus four recurrent combinations, or the same endpoint gains plus four unrecorded ordinary gains.

The recording comparison retained the four endpoints and selected a fifth site from the remaining neurons. One policy used a fixed random site, one minimized uncertainty in the recurrent combinations, and one minimized the existing prior-averaged response risk. A four-endpoint allocation measured the effect of adding the fifth observation. Selection used the public model and prior, before generating calibration observations. All policies shared one physiological draw and one noise draw per network; identical sets reused the same fit.

For the combination criterion, endpoint columns were removed from the four rows defining $R\,\mathrm{diag}(g)L$. An orthonormal basis B for the remaining row space expressed four internal combinations in prior-standardized coordinates. The criterion averaged $\mathrm{tr}(BC_SB^T)/4$ over the same 32 prior particles used by the response criterion, where $C_S = (I + F_S^T F_S)^{-1}$. This is a local A-optimal objective for the specified combinations. The response comparator retained the same four endpoints and chose its fifth site using the original three-response objective. Combination estimation error was the mean square of $B(\hat{z} - z_{\mathrm{true}})$.

One recurrent direction was fixed before observing calibration data: the leading eigenvector of the prior-averaged combination covariance after four-endpoint recording, mapped back through B. The same direction was profiled for every allocation. Its minimum and maximum were sought in the nonlinear MAP-objective sublevel set with allowance 2, using two starts per sign, invertible Gauss–Newton coordinates and a 120-iteration cap. These numerical feasible widths assess one specified direction; they are not confidence intervals for the full combination space. Local covariance traces and estimation errors supplied separate diagnostics.

The three primary comparisons were response MSE in the oracle contrast, squared profile width for combination-targeted versus random fifth-site recording, and response MSE for that same recording contrast. Summaries include paired-network 95% bootstrap intervals and Bonferroni intervals across these three comparisons. Comparisons with response-based selection, combination estimation errors and four-versus-five recording contrasts were secondary. Nine common starts were used initially. An unsuccessful selected fit, an objective above the known feasible generating-parameter objective, or a lower objective found during profiling triggered nine further fits for every policy: its own best or profile point and eight common prior draws. At most two such rounds were allowed. Oracle information was never transferred into the recording policies.

5.14 Independent fifth-recording follow-up

A new cohort of 80 networks (61000-61079) tests response-directed fifth-site selection against a random fifth site as its single primary comparison. The model and the four recording policies are unchanged from the preceding study. Each network has one physiological draw, three independent paired observation-noise draws and the same three response targets. Recording sets are selected from the public model and 32 prior particles before calibration observations are generated. Selection is reused across the three noise draws; identical sets share fits.

The response MSE is averaged across targets and noise draws within each network. Percentage changes and 95% intervals use 30,000 paired-network bootstrap resamples. The sample size, main comparison, source code and analysis were locally fixed before validation outcomes. Comparisons with four-endpoint recording and combination-directed selection are secondary. All fits begin from nine common starts with a 400-evaluation cap. A selected fit that fails to converge or exceeds the known feasible generating-parameter objective triggers a matched additional round for every distinct policy, using its best point and eight new common prior starts, up to two rounds. Generating parameters are used for evaluation and this feasible-cost check.

A shared-reference diagnostic separates the effects of the additional observations from changes in the fitted reference point. At each four-endpoint MAP fit, the observation Jacobians and response gradients are evaluated once. For every recording set, the local covariance $C_M = (I + F_M^T F_M)^{-1}$ is formed at this same parameter vector. The response gradient is split into its projections onto endpoint coordinates, the orthonormal internal-combination space and the remaining orthogonal space. The three diagonal terms and all cross terms sum exactly to $\mathrm{tr}(GC_MG^T)/3$. The internal-information diagnostic is $\mathrm{tr}(BC_MB^T)/4$. Along the four eigenvectors of the internal covariance after endpoint recording, central finite response changes at standardized parameter lengths 0.1 and 0.3 are compared with the local sensitivities. These

diagnostics were specified with the validation protocol; they assess the local uncertainty approximation separately from realised prediction error.

An exploratory finite-error decomposition evaluates all eight combinations of generating and fitted parameter blocks. The blocks are endpoint coordinates, the internal-combination projection and its orthogonal remainder. Averaging each block's response increment across its possible orders gives three contributions that sum exactly to the fitted-minus-generating response. Their squared terms and all pairwise cross terms reconstruct the response MSE. This algebraic attribution describes compensation among parameter errors. In the repeated-noise cohort, the sample prediction variance and the squared error of the mean prediction, corrected for its sampling variance, also separate observation-noise variability from noise-averaged prediction bias. These exploratory analyses use every retained network and do not affect selection or the primary comparison.

5.15 Study accounting and availability

The companion package contains the independent implementation, dependency lock, actual execution/source freezes, per-instance outputs, numerical controls, network-level analysis and figure source data. MIASE, DOME and MDAR inform reporting [37–39]. No new human or animal measurements were collected. Source equations were independently implemented; unlicensed upstream code is not redistributed. All freezes were local records rather than external preregistrations.

The response-range and indirect-recording studies each add forty independent 100-unit networks. The parameter-combination/structure study and limited-budget study add two further forty-network cohorts. A final forty-network cohort supplies the dimension-matched information confirmation and recording-acquisition comparison. The independent fifth-recording follow-up adds eighty networks with three noise repeats each. The 100-unit total is 368 independent base networks: eight pilot networks, seven disjoint forty-network cohorts and this eighty-network cohort. Structural variants are paired within their forty base seeds. Retrospective factor decomposition reuses the response-range cohort; the first combination interventions and fitted-versus-true decomposition share the structure cohort. The forty-network information and recording comparisons share their cohort and are counted once. The eighty-network follow-up has a separate primary comparison. Cohorts are not pooled across differing estimators or experimental questions.

6 Data and code availability

The original frozen analysis tables and figure source data remain in the collaborator's research package. The repository workspace `work/recording_extensions` contains the independent confirmation and matched-width experiments; its `round2` directory contains the endpoint cohorts, common fitting follow-up, source code, protocols, parameter-level outputs and analysis tables. The round3 directory adds the two new independent cohorts, locally frozen protocols, full numerical response searches, gain-information interventions, indirect-recording criteria, numerical audits and source figures. Its round4 directory contains the finite-response decomposition, parameter-combination interventions, structural robustness and limited-budget allocation studies. The round5 directory contains the independent information confirmation and the recording-acquisition comparison, including nonlinear combination profiles. The round6 directory contains the independent eighty-network fifth-recording follow-up, its fixed protocol, repeated-noise fits, shared-reference uncertainty diagnostics and exploratory finite-error decompositions. The same workspace contains the extensions, their dependency lock, and integrity checks. Upstream code and weights from the published model family are not redistributed.

7 Author contributions

J.D. and S.C. contributed equally to this work, including conceptualisation, methodology, investigation, visualisation and writing.

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Supplementary analyses and figures

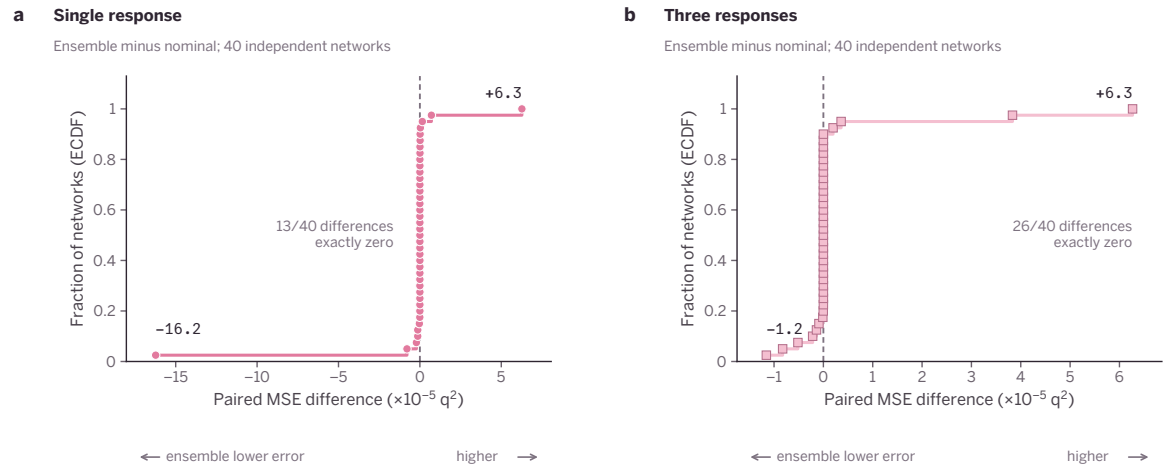


Fig. S1 | Complete paired primary differences. Each panel shows all 40 independent network-level ensemble-minus-nominal raw-MSE differences for the single-response (a) and three-response (b) comparison, including exact ties and the influential networks; the step curve is the empirical distribution function and every sorted value is drawn as a marker. The horizontal axis is in units of 10^{-5} q^2; negative values favour the ensemble. Extreme values (-16.2 and $+6.3$ in a; -1.2 and $+6.3$ in b) are labelled, not truncated. Exactly 13 (a) and 26 (b) of the 40 differences are zero. The distributions are descriptive; no interval is drawn, and the primary intervals remain those in Supplementary Table 1 and Supplementary Fig. 12. No equivalence margin was specified, and nothing here shows that the two selection rules perform alike.

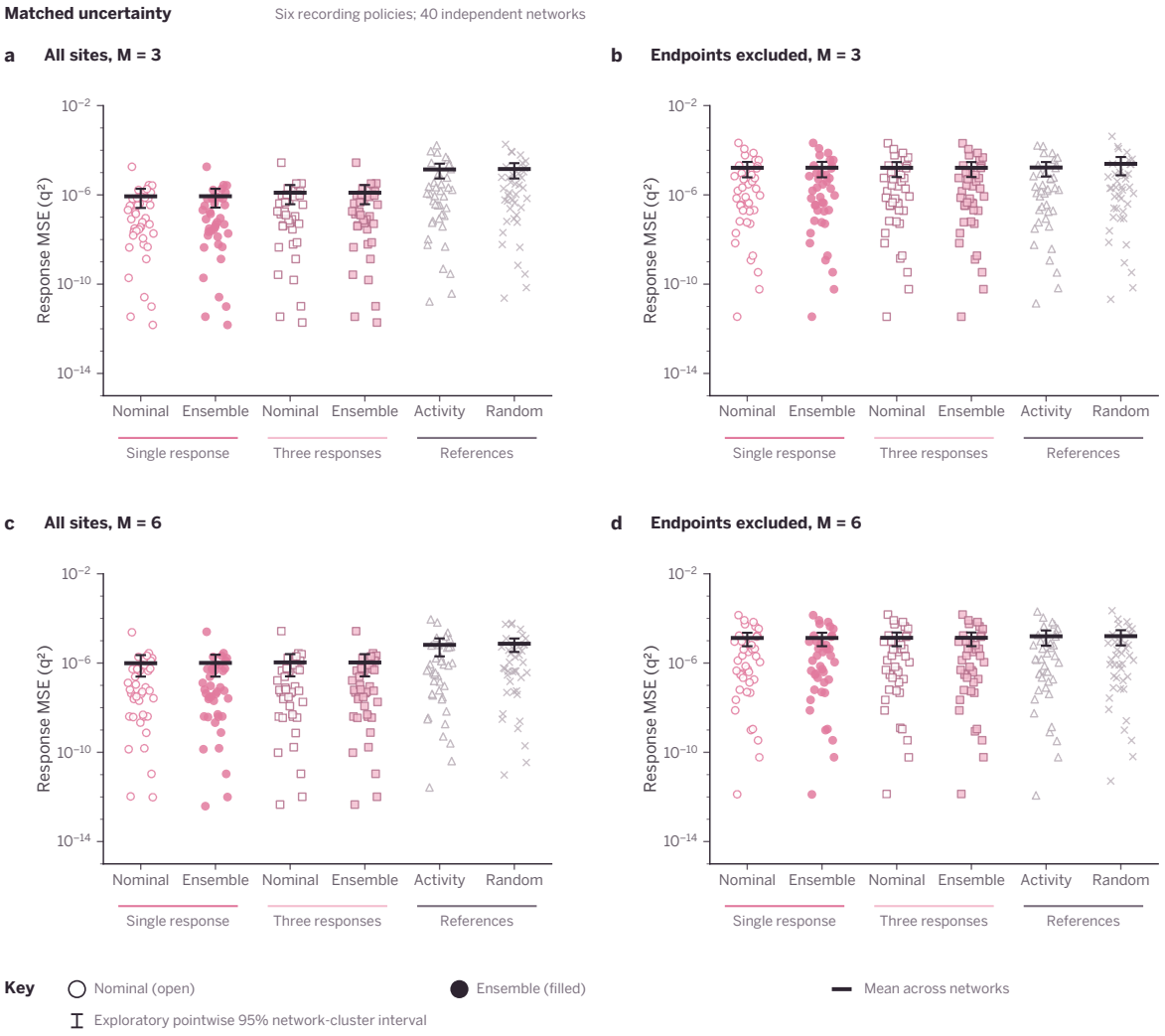


Fig. S2 | All recording conditions under matched uncertainty. Panels a–d show all-site/M = 3, endpoint-excluded/M = 3, all-site/M = 6 and endpoint-excluded/M = 6. For each of six policies (nominal and ensemble single-response, nominal and ensemble three-response, activity, random) every one of the 40 independent networks is shown as one point, with truth, noise and readout repeats averaged within network. Thick black bars are the aggregate means and thin black lines the exploratory pointwise 95% network-cluster intervals from the frozen summary table. Open markers denote nominal and filled markers ensemble selection; horizontal jitter only separates overlapping points. All panels share the logarithmic response-MSE axis (10^{-15} to 10^{-2}). No point or tail is removed. Intervals are not multiplicity-adjusted and no pairwise or equivalence claim is made.

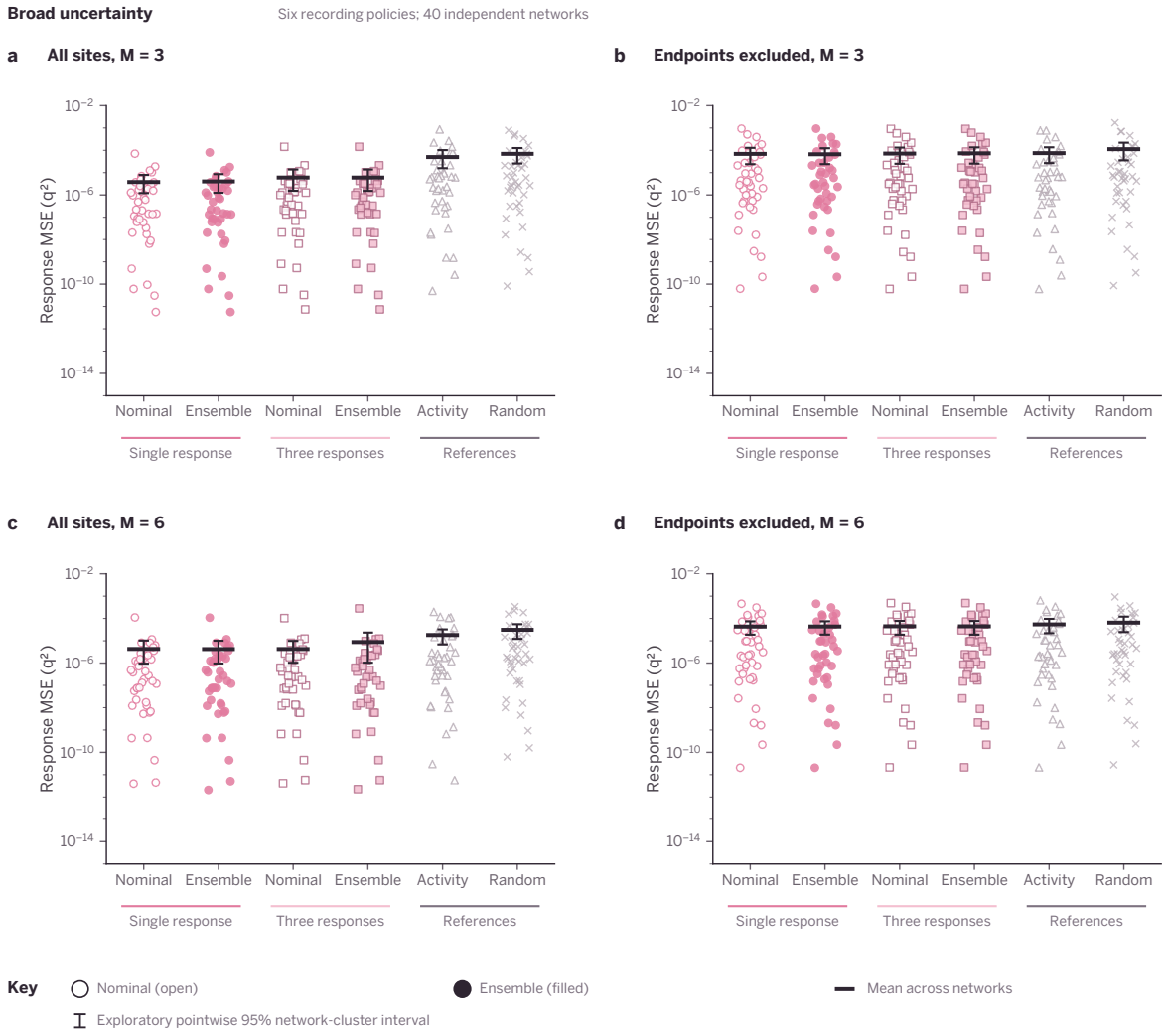


Fig. S3 | All recording conditions under broad uncertainty. As Supplementary Fig. 2 for the broad prior: panels a–d show all-site/M = 3, endpoint-excluded/M = 3, all-site/M = 6 and endpoint-excluded/M = 6; 40 independent networks per policy and panel; black bars and lines are aggregate means and exploratory pointwise 95% network-cluster intervals; shared logarithmic axes; open nominal, filled ensemble; jitter is horizontal only. The primary comparison in Supplementary Fig. 12 corresponds to panel b. No method-specific tail is removed, and the panels do not support any claim of equivalence or superiority.

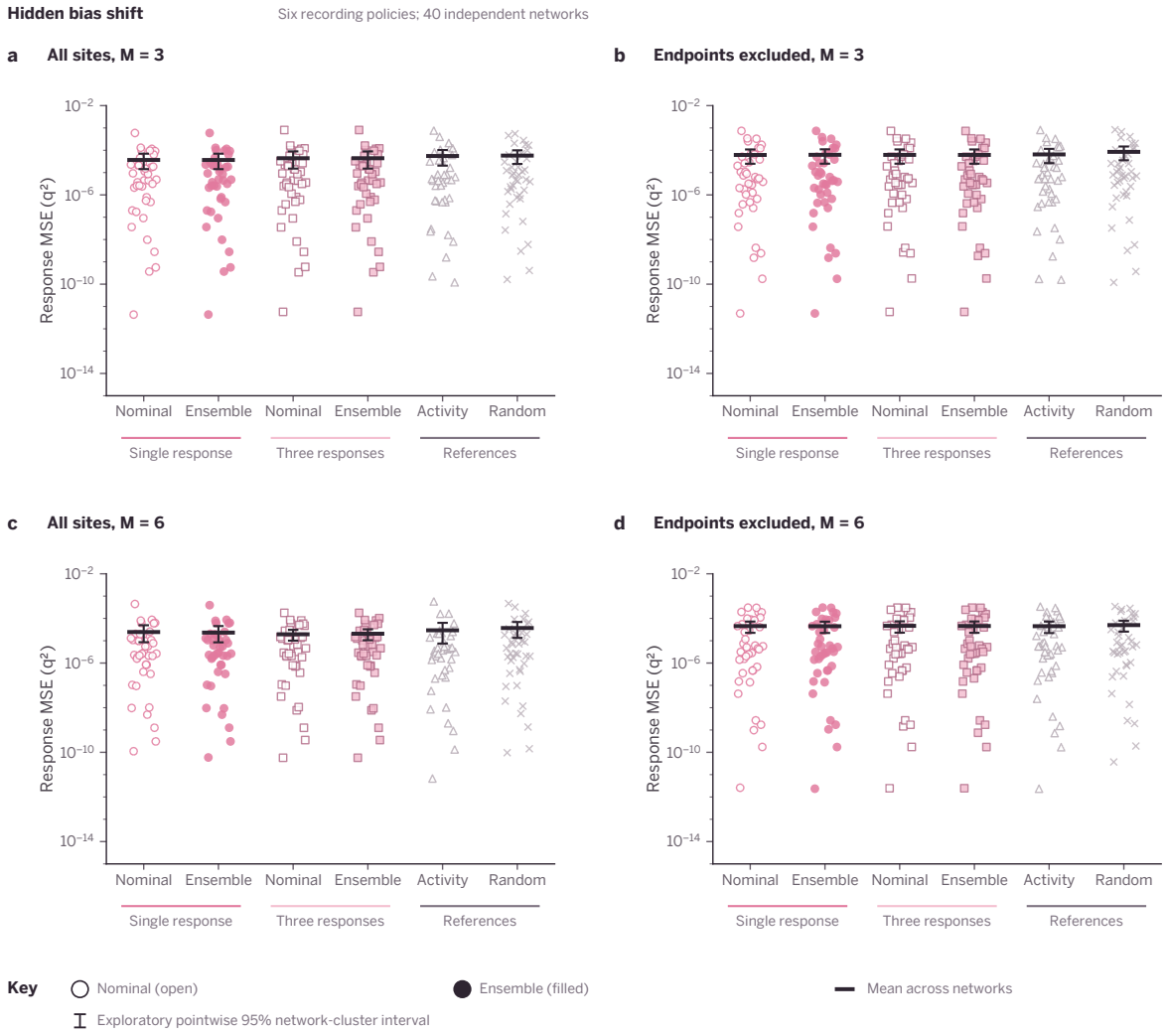


Fig. S4 | All recording conditions under hidden bias shift. As Supplementary Fig. 2 for the hidden-bias-shift condition: panels a–d show all-site/M = 3, endpoint-excluded/M = 3, all-site/M = 6 and endpoint-excluded/M = 6; 40 independent networks per policy and panel; black bars and lines are aggregate means and exploratory pointwise 95% network-cluster intervals; shared logarithmic axes; open nominal, filled ensemble; jitter is horizontal only. No data are filtered, and the intervals are descriptive and not multiplicity-adjusted.

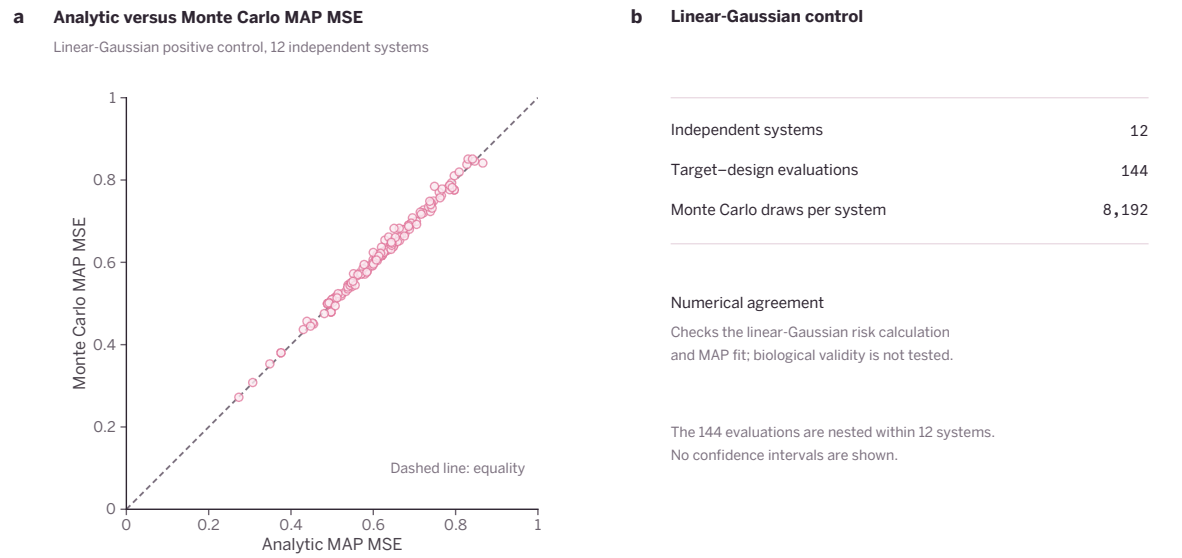


Fig. S5 | Nondegenerate linear-Gaussian positive control. All 144 target/design evaluations from 12 independent linear systems compare the analytic MAP MSE (horizontal) with the Monte Carlo MSE from 8,192 draws per system (vertical); the dashed identity line is the analytic reference. The independent unit is the system (n = 12); the 144 points are evaluations within systems, not independent neural networks, and no confidence intervals are displayed. Agreement validates this computational special case only; it does not validate biological assumptions.

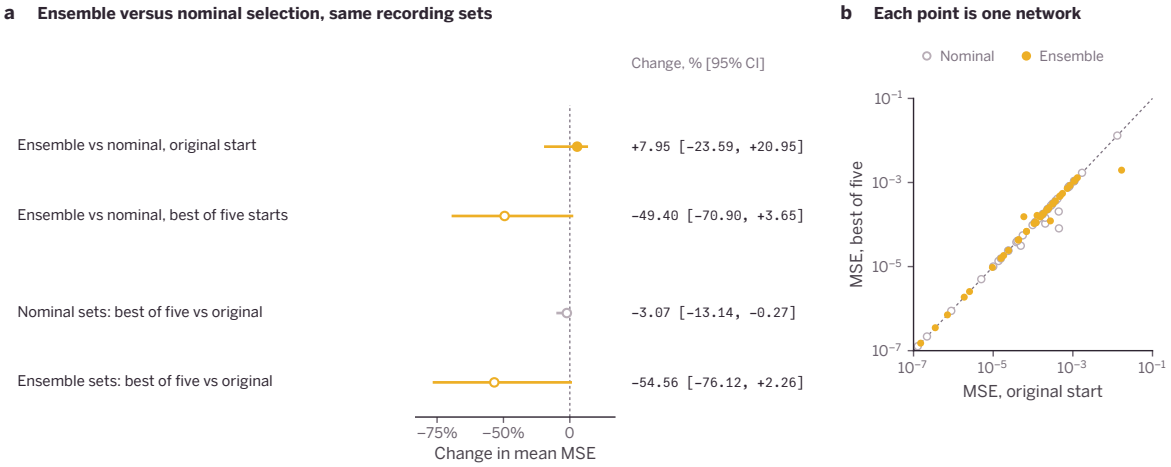


Fig. S6 | Fitting sensitivity in the primary 100-unit condition. Recording sets are fixed; both selectors receive the same five-start rule and retain the smallest MAP objective. **a,** Changes in mean response MSE for ensemble versus nominal selection with the original start and with the best of five starts, and for each selector's best-of-five fit relative to its original fit, with paired 95% network-bootstrap intervals. The original-start contrast is the frozen confirmation comparison (filled marker). **b,** Network mean MSE from the original start and the best of five starts for nominal (open) and ensemble (filled) selection; the dashed line marks equality. Forty previously evaluated networks; retrospective sensitivity analysis.

Additional indirect-recording comparison

The held-out-prior versus nominal response-MSE change was -3.48% (95% CI -34.14% to +26.28%). The prespecified comparison did not establish an improvement from the held-out-prior criterion. The internal covariance criterion versus nominal gave -22.77% (95% CI -52.62% to -5.70%); prior-averaged local risk versus nominal gave -13.56% (95% CI -46.73% to -3.90%). All policies used the same budget and inference rule. The covariance-versus-ensemble comparison gave -10.65% (95% CI -18.73% to +3.93%); covariance versus random recording gave -5.50% (95% CI -24.37% to +1.72%). In the retrospective optimization sensitivity check, covariance versus nominal gave -11.50% (95% CI -17.17% to +15.04%). These tests concern an application of established covariance and predictive-validation ideas, not a new general experimental-design principle.

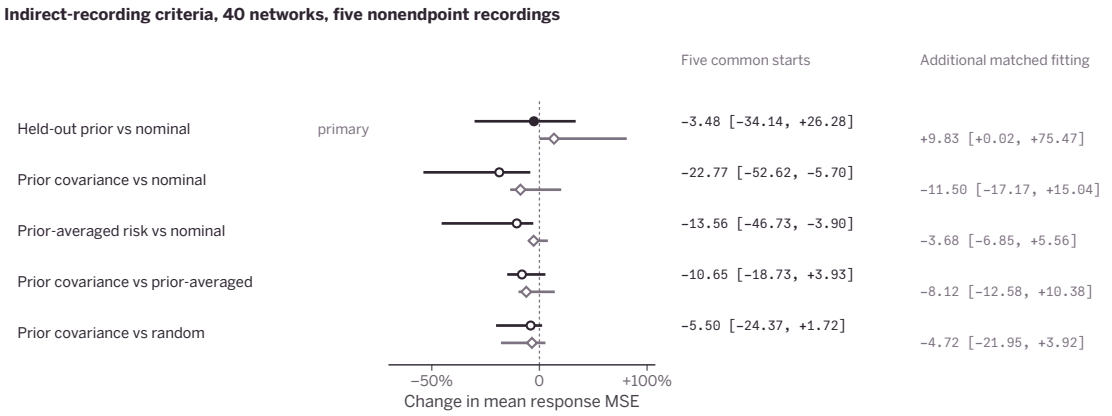


Fig. S7 | Indirect-recording criteria on forty independent networks. Points and intervals show paired mean-MSE changes and 95% network-bootstrap intervals. Circles use five common starts; diamonds include additional matched fitting for all policies on eleven flagged datasets. The original held-out-prior comparison is primary (filled marker); other comparisons and the fitting sensitivity check are exploratory.

Limited-budget recording comparison

A separate forty-network study tested budgets two and three under uniform noise, fourfold stimulus-site noise SD, and fourfold readout-site noise SD. Endpoint-pair lookahead selected exactly the same sites as ordinary ensemble greedy selection in 237/240 paired settings. Its prespecified aggregate MSE change was -0.00136% (95% CI -0.00555% to 0.00000%), averaging the six settings within each network first. Relative to exact endpoint-only subset enumeration, its change was +2.54% (95% CI -0.11% to +7.68%). The expanded search therefore supplied negligible additional predictive benefit over ensemble greedy selection (Supplementary Fig. 8).

The simple endpoint-only search itself changed mean MSE by -46.29% (95% CI -59.65% to -21.95%) relative to recording the stimulation site first and filling the remaining slots with randomly ordered readouts, in a secondary comparison. Thus the choice among accessible endpoints matters under a restricted budget. The existing ensemble-risk objective already captured most choices supplied by pair seeding in this cohort; these results support a useful endpoint baseline rather than a distinct advantage from the added search procedure.

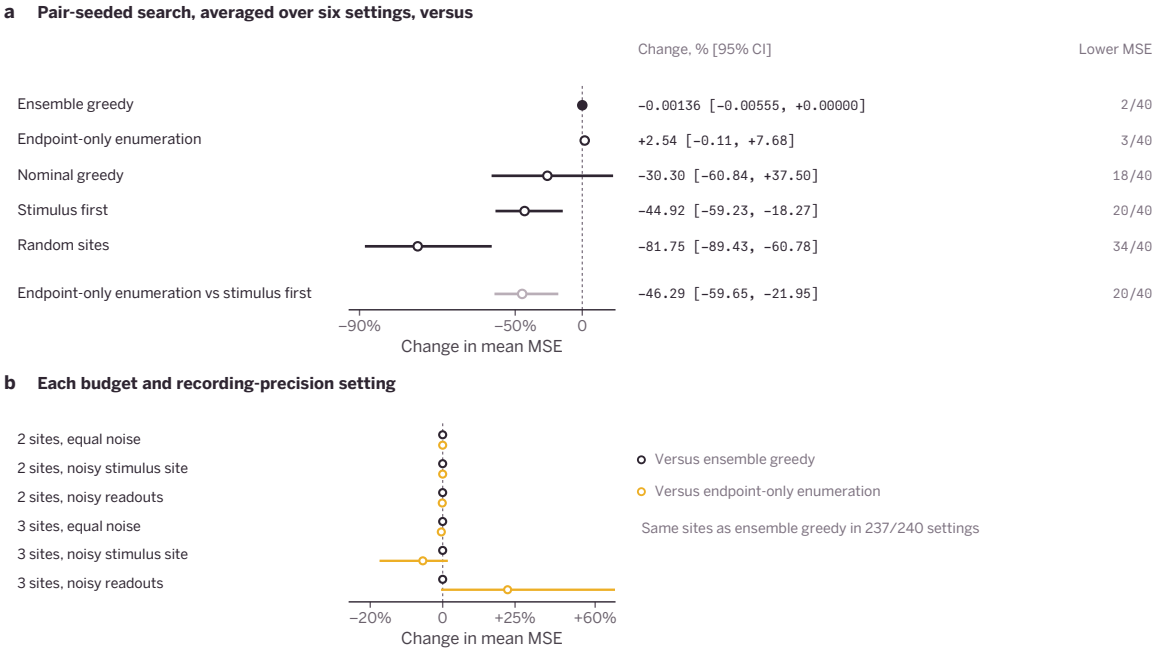


Fig. S8 | Endpoint-pair search under limited budgets and unequal precision. a, Changes in aggregate mean MSE for pair-seeded selection relative to five alternatives, after averaging equally over two budgets and three noise settings within each of forty independent networks; the bottom row compares exact endpoint-only enumeration with stimulus-first recording. The right column counts networks with lower MSE. **b,** Separate settings compare the candidate with ordinary ensemble greedy selection and exact endpoint-only subset enumeration. The aggregate comparison against ensemble greedy is primary; individual settings and other comparisons are secondary. Identical recording sets reuse the same observations and fitted solution. Intervals are pointwise 95% paired network-bootstrap intervals. All selected fits terminated successfully and passed the known-feasible objective check. Shared prior-bank construction is separate from selection: mean bank time was 18.89 s, while mean search times were 5.57 s for pair lookahead, 2.70 s for ensemble greedy, and 0.048 s for endpoint-only enumeration. These descriptive times reflect the implementation and concurrent workstation workload.

Static combination controls

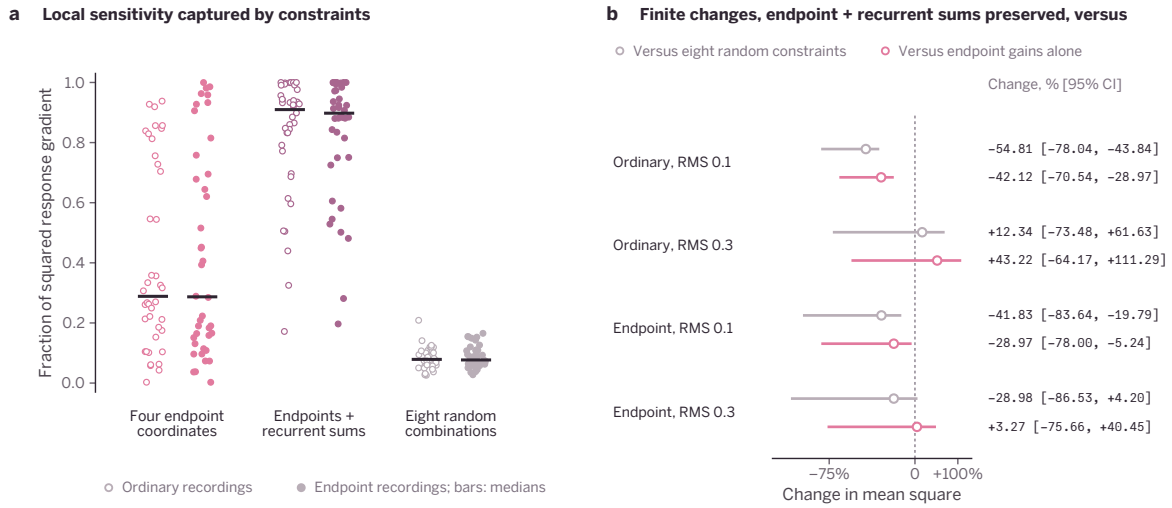


Fig. S9 | Local sensitivity and finite changes preserving gain combinations. a, Fraction of squared response-gradient norm projected onto four endpoint coordinates, endpoint coordinates plus four recurrent combinations, or eight random linear combinations, under ordinary (open) and endpoint (filled) recordings. Dots are independent networks and horizontal marks are medians. **b,** Mean squared response changes after equal-norm parameter perturbations preserve the eight endpoint/recurrent combinations, relative to perturbations preserving eight random combinations or endpoint gains alone. Parameter RMS changes are expressed in prior standard deviations. All forty new networks and both perturbation sizes are retained. Intervals resample networks at the 95% level; the interventions preserve the specified parameter combinations and do not impose a calibration-fit constraint.

Original criterion-comparison studies

Original criterion-comparison model

The original 12-unit model is

$$\dot{x} = -x + Jr + u, \qquad r_i = \exp(\eta_i)\,\mathrm{softplus}(x_i + b_i). \tag{10}$$

N=12, x(0)=0 and J is known. The independent synthetic J matrices have 30% sparse connectivity, zero diagonal, column-wise excitatory/inhibitory signs, and spectral norm 0.65. The uncertain vector is $\theta=(\eta,\mathrm{b})$. The observation is the rate r, not the internal state x. Time and activity are dimensionless.

Two known calibration inputs apply sinusoidal probes to three channels. Euler integration uses a step of 0.05 over six time units. Sampling every fourth step yields 60 observations per recorded neuron across the two calibration trials. Measurement noise is independent Gaussian with standard deviation 0.05.

A separate fixed stimulus of amplitude 0.6 acts during time 1–2. Three readouts are selected from structurally reachable nodes using J before physiological parameters or measurement noise are generated. The target family is

$$q_k(\theta) = \int_0^6 [r_{j_k}(t;\theta,u_*) - r_{j_k}(t;\theta,0)]dt. \tag{11}$$

The primary implementation evaluates this integral by the matching step-weighted (left-point) sum with dt=0.05. Q consists of these three responses. The vector criterion and primary raw-error evaluation use equal weights over the same target family. Scalar designs specialise for one known target; vector designs use one configuration for all three. No observations from separately selected scalar configurations are pooled.

Recording budgets are M=3 and M=6. The candidate pool either includes all neurons or excludes the stimulation site and all three readouts. The latter pool is shared by every target and method.

Development and the first frozen evaluation

Stage	Independent J	Physiological truths/J	Noise repeats/truth	Readouts/J	Conditions
Development	8	2	2	3	Matched, broad, bias shift, gain-only, bias-only
First frozen evaluation	40	3	3	3	Matched, broad, bias shift

Matched and broad conditions use correctly specified Gaussian priors and generating standard deviations 0.18 and 0.35 respectively. Bias shift retains the design prior at 0.18 while adding 0.5 to the hidden true bias mean. The method is not informed of this shift. Development gain-only and bias-only conditions fix the other parameter group. Increasing a prior scale changes parameter variability, signal scale and nonlinearity together; it is not an isolated curvature intervention.

Connectivity, truth, noise, design ensembles, task-scale Monte Carlo, random sets and analysis use separate seed namespaces. The first evaluation uses network IDs 5000–5039, disjoint from development. The execution protocol and computational source were locally frozen before outcomes were inspected. This is not external preregistration.

Original-study estimands and uncertainty

For each J, squared response errors are averaged across all prescribed truths, noises, targets and random-design repetitions. Network risks are then averaged equally across J. RMSE is the square root of the overall mean squared error. Repeated targets or noises do not create additional independent networks.

Interval key: 97.5% intervals belong to the two prespecified first-study comparisons, the 95% interval to the single prespecified forecast comparison, and other intervals are pointwise 95% unless specified otherwise in the follow-up protocols. The six structural contrasts additionally report six-comparison Bonferroni intervals. Two prespecified primary comparisons, ES−LS and EV−LV, concern broad uncertainty, excluded endpoints and M=3. The paired network-cluster percentile bootstrap [40, 41] uses 10,000

resamples of whole networks, so repeated targets and noises do not count as independent. Each primary comparison receives a 97.5% interval as a two-comparison Bonferroni interval arrangement. Other strata, task-specialisation comparisons and factorial interactions have exploratory pointwise 95% intervals. We do not infer equivalence from an interval crossing zero [32, 33].

The primary scale is raw response MSE. A secondary scale uses an independent common 64-draw nonlinear prior-Monte-Carlo variance. It is not method-specific. Prescribed targets below a frozen variance floor of 10^{-16} retain raw results; a complete standardised estimand is NA rather than silently excluding them. Local MAP intervals are approximate diagnostics, not guaranteed confidence intervals.

Separately specified risk-forecast follow-up

The first study distinguishes selection gains from apparent risk-estimation differences only exploratorily. A subsequent protocol therefore uses new network IDs 6000–6039 and fixes the recording policies to LS and LV. The point predictions and selected data are identical for the two risk forecasts being compared. The forecasts are the nominal and ensemble local risks evaluated at that same recording set.

The risk-forecast score is

$$S(v, e) = \log v + e^2/v, \qquad v = \max(\hat{R}, 10^{-24}). \tag{12}$$

For positive v, its conditional expectation is minimised at v=E[e^2]. This makes it a consistent score for a positive MSE forecast; it does not establish a full predictive distribution or require the errors to be Gaussian. The log term penalises indiscriminate inflation. This is a use of established scoring principles [42, 43], not a new score. The forecast-evaluation literature distinguishes calibration from sharpness [44] and treats calibration as separate from point-prediction accuracy [45]; the nominal-versus-ensemble comparison here mixes both, applies to a mean-squared-error forecast rather than a full predictive distribution, and does not test calibration separately.

The follow-up has one frozen primary comparison: ensemble minus nominal risk-forecast score for the LS recording policy under broad uncertainty, excluded endpoints and M=3. It uses a paired network-cluster 95% interval. Matched/shift conditions and LV are secondary. These results do not redefine the first study's primary hypotheses.

The original 12-unit studies and the subsequent 100-unit criterion confirmation preceded the endpoint and mechanism experiments. Their frozen primary comparisons, all retained conditions and fitting diagnostics are reported here. Each study retains its original evaluation population and primary comparison.

Original criterion-comparison setup

The original 12-unit networks have known numerical connectivity J and uncertain neuronal gains and biases (symbols in the Notation table, Methods). Calibration data were generated from a limited set of nodes; the evaluation target was the time-integrated response to a separate fixed stimulus, relative to an unstimulated control (Supplementary Fig. 11). Three readouts were chosen using structural reachability before generating physiological parameters or measurement noise. The scalar objective specialised its recording set for one readout, whereas the vector objective used one set for the same three-response family. This construction separates task specialisation from the comparison between nominal and ensemble risk. Throughout the Results, "single-response" (scalar) and "shared" (three-response, vector) name the two objectives; the abbreviations LS, ES, LV and EV, defined in Methods, combine objective and risk criterion.

Box 1 and Supplementary Fig. 10 summarise the audit. All methods fitted the selected observations with the same Gaussian MAP estimator and predicted the response by nonlinear simulation. The principal candidate pool excluded the stimulation site and all three readouts, preventing direct endpoint recording. All-site recording and a larger budget were retained as additional conditions. Exact dataset reuse avoided redundant fits when methods selected identical nodes; no observations from different scalar designs were pooled. Box 1 (below) defines the three-step comparison used in these original studies: Decision, Error and Forecast.

Box 1 | Three-step audit of a recording criterion. Inputs: a design pool, a budget M, a stimulus and target q, an estimator, and two criteria A (nominal) and B (prior-averaged). 1. *Decision.* Select a recording set with each criterion. If the sets are identical the fitted data, predictions and errors

are identical, so only the remaining designs can differ. 2. *Error.* Fit with the common estimator and compare response error against hidden truth across all designs, including those with identical recording sets, paired by independent network, with a cluster-bootstrap interval. 3. *Forecast.* Fix the recording set, data and point prediction; compare the two risk forecasts of the error with a score that is minimised at the true mean squared error, again paired by network.

Report agreement counts, intervals read as approximate bounds, and the influence of the largest single network.

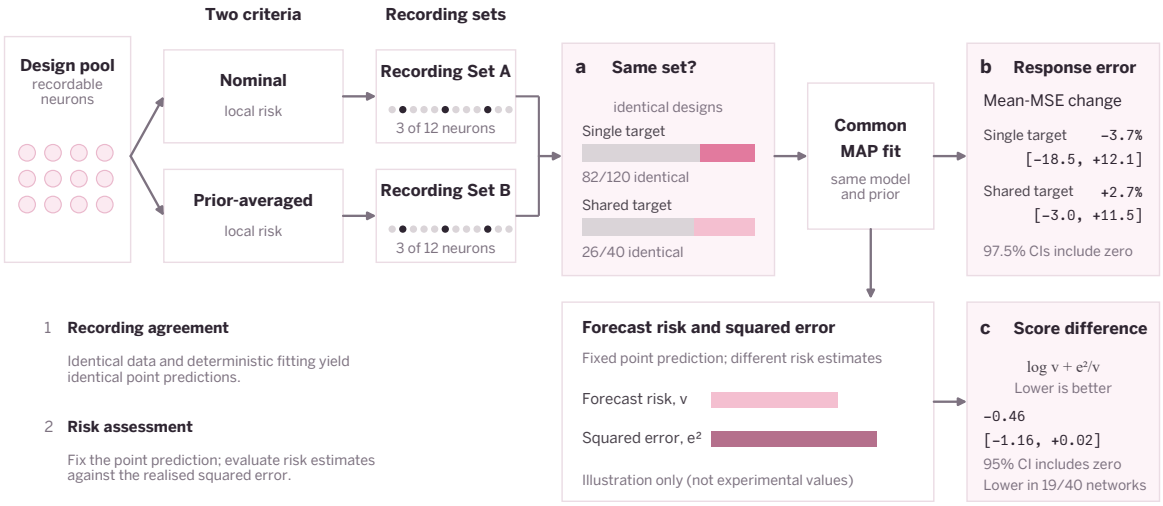


Fig. S10 | The three-step audit. Two criteria, nominal and prior-averaged local risk, each select a recording set from the same design pool. Panel a asks whether the two sets are identical; if they are, the common maximum a posteriori fit sees the same data and the response error cannot differ. Panel b compares the response error of the fitted prediction against hidden truth (mean-MSE change with the 97.5% interval for the single-response and shared objectives; both intervals include zero). Panel c fixes the data and the point prediction and compares the two risk forecasts v of the error e with the score $\log v + e^2/v$. The reported mean score difference is prior-averaged minus nominal (95% interval includes zero). The counts, effects and intervals are those of Supplementary Figs. 12–13 and Supplementary Table 1; the dot and bar icons for the recording sets and for v and e^2 are illustrative and carry no data. Negative differences favour the prior-averaged criterion.

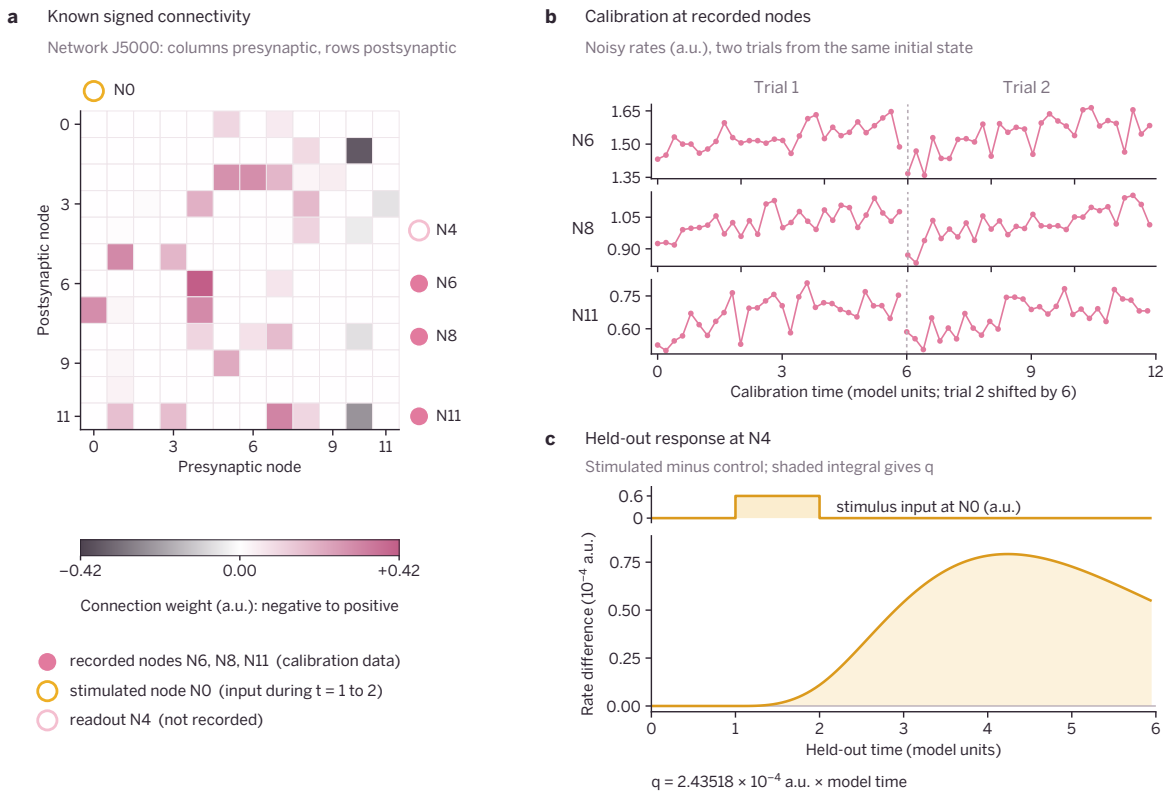


Fig. S11 | Connectivity, limited recordings and a held-out response in one simulated network. a, Known signed connectivity J for the first frozen network, J5000 (rose positive, deep plum negative weights; symmetric colour limits ±0.42 around exact zero); columns are presynaptic and rows postsynaptic nodes. Filled markers beside the matrix identify the recorded nodes N6, N8 and N11, the open light-rose marker the unrecorded readout N4, and the open amber marker the stimulated node N0. **b,** Actual noisy calibration observations at the three recorded nodes in two trials, each reset to the same known initial state (Gaussian observation noise SD 0.05, stride 4). Trial 2 is shifted by 6 model-time units for display; the dashed line marks the trial boundary, not continuous evolution, and lines connect observations only within a trial. Vertical scales differ between nodes. **c,** Stimulated minus matched zero-input control rate at N4 (displayed in units of 10^{-4}); the upper strip shows the stimulus input at N0 during model time 1 to 2. The integrated response $q = 2.43518 \times 10^{-4}$ (a.u. × model time) is Δt times the sum of the 120 frozen-model differences ($\Delta t = 0.05$), not a new trapezoidal integral. The case is the fixed first-index instance (broad uncertainty, truth and noise draw 0, target 0, endpoints excluded, $M = 3$, nominal single-response selection), chosen before its effect was inspected. Independent $n = 1$ network; no interval is shown and no between-method effect or biological validity is claimed. All quantities are in dimensionless model units.

Before the methods were compared, the code was checked against an independent implementation and a linear-Gaussian case with a known answer. An independent rate/sensitivity implementation agreed within approximately 3.4×10^{-16}, and finite differences supported the analytic derivatives. A nondegenerate linear-Gaussian positive control gave identical nominal and ensemble designs in all 48 design pairs; analytic prediction risk agreed with Monte Carlo MAP error across 144 target/design evaluations (Supplementary Fig. 5). A separate additive-bias control produced the expected zero derivative of the paired stimulus response. These checks validate specific computational operations, rather than biological assumptions.

Prior averaging often selects the same recordings and did not establish lower error in the primary comparisons

The Decision and Error steps of Box 1 apply to the first frozen evaluation, which used 40 independent J matrices, with three physiological truths, three noise repetitions and three readouts nested within each network. The primary condition combined broad matched uncertainty, excluded endpoints and a budget of three recorded nodes. For each method, errors were first averaged within J; mean risks and paired intervals were then calculated across networks. All fits terminated successfully, with no fallback or invalid task scale.

Table S1 | Frozen primary response-error comparisons

Objective	Nominal RMSE	Ensemble RMSE	Mean MSE difference, E−N	97.5% interval	Relative mean-MSE change [97.5% interval]
Single response	0.008312	0.008155	-2.586×10^{-6}	$[-1.471\times10^{-5}, 4.826\times10^{-6}]$	−3.74% [−18.48%, 12.12%]
Three responses	0.008451	0.008564	1.924×10^{-6}	$[-1.148\times10^{-6}, 6.826\times10^{-6}]$	2.69% [−2.95%, 11.51%]

Broad matched uncertainty; endpoints excluded; M=3; 40 independent networks. RMSE is the square root of the overall mean network MSE. The primary estimand is the raw paired mean-MSE difference. The relative effect is a ratio of aggregate means minus one, not an average of individual percentage changes. Each prespecified comparison uses a 97.5% paired network-cluster interval.

The two criteria often chose the same recordings (Supplementary Fig. 12a). A descriptive readback of the frozen designs found identical recording sets in 82 of 120 target-specific designs (68%; three designs per network) and in 26 of 40 shared designs (65%). Because identical sets give identical fitted data and predictions by construction, a difference in error can arise only in the remaining designs. At the network level, the ensemble and nominal MSE were exactly equal in 39 of 80 network–objective cells (13 single-response and 26 three-response networks; Supplementary Fig. 12b). This count is descriptive and was not a prespecified hypothesis.

The ensemble scalar design had a slightly lower mean MSE than its nominal counterpart, but the paired interval included both improvement and deterioration (Supplementary Table 1; Supplementary Fig. 12c). The vector comparison had the opposite point estimate and was also uncertain. Relative mean-MSE changes were −3.74% for ES versus LS and +2.69% for EV versus LV. A leave-one-network diagnostic changed the sign of the scalar mean difference when one influential network was omitted; that network remains in every primary estimate. The results do not establish a stable ensemble advantage or method equivalence.

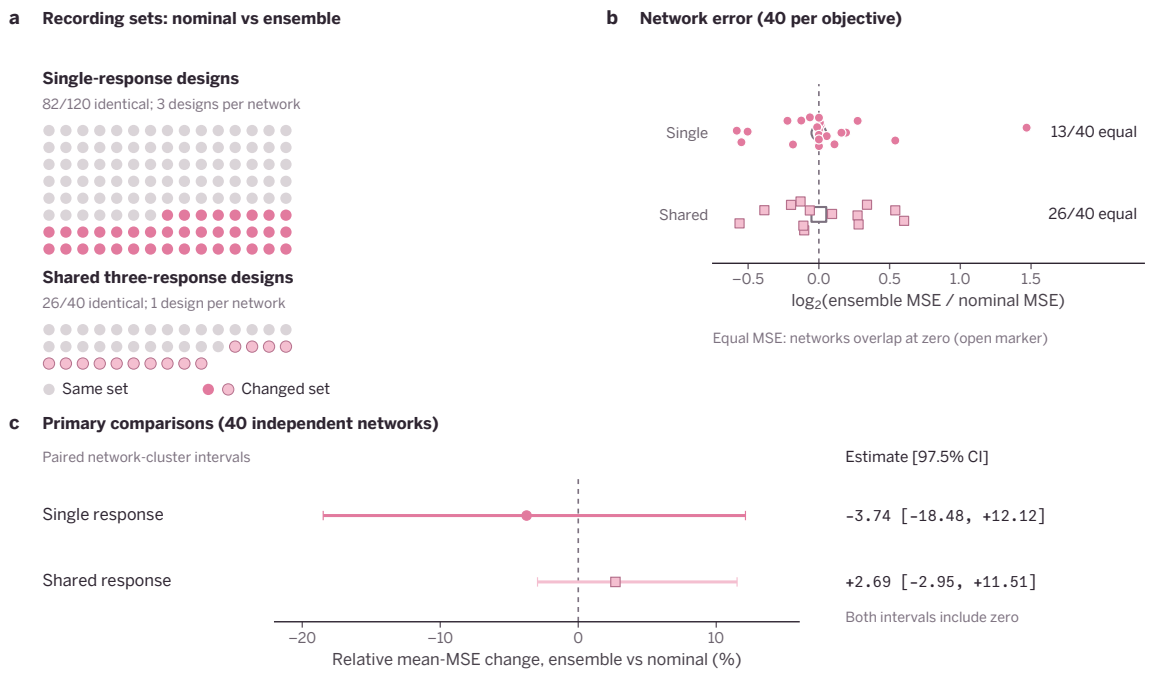


Fig. S12 | Prior averaging often selects the same recordings; primary error comparisons remain uncertain. **a,** Each dot is one recording design from the frozen first evaluation (120 single-response designs, three per network; 40 shared designs). Grey dots are designs in which the nominal and ensemble criteria chose identical recording sets; coloured dots are designs in which they differed (82/120 and 26/40 identical). **b,** Descriptive within-network $\log_2$ ratio of ensemble to nominal MSE for the 40 independent networks per objective; the hollow marker at zero stands for networks with exactly equal MSE (13 single-response, 26 three-response), and filled points are the remaining networks. Three truths, three noise repetitions and three readouts are averaged within each network. These ratios are neither the primary estimand nor the ratio of aggregate means. **c,** Relative change in aggregate mean MSE, ensemble versus nominal, with the prespecified 97.5% paired network-cluster bootstrap intervals (same estimates as Supplementary Table 1). Negative values favour the ensemble. All networks, including influential instances, are retained; Supplementary Fig. 1 shows the complete raw difference distributions. No method-equivalence margin was specified.

Recording constraints changed the achievable error. For nominal scalar design at M=3, RMSE was 0.000929 with all sites available and 0.004046 with endpoints excluded under matched uncertainty; the corresponding broad-condition values were 0.001946 and 0.008312. These exploratory comparisons make direct endpoint access an important part of the task definition. The activity and random references, both budgets, and all uncertainty conditions are retained in Supplementary Figs. 2–4.

A separately specified forecast test separates risk estimation from point-prediction accuracy

In the first study, an exploratory aggregate comparison suggested that ensemble risk could match actual error better than nominal risk does, even when it did not improve the selected recordings. At the primary nominal scalar recording set, actual MSE divided by nominal forecast risk was approximately 1.615; the ratio using ensemble forecast risk was 1.103. Such a ratio does not establish forecast accuracy across networks. The Forecast step of Box 1 therefore became a separate test, specified by us, using 40 new J matrices, fixing the recording policy, selected data and fitted point predictions while comparing the two risk forecasts.

The primary ensemble-minus-nominal MSE-forecast score difference was −0.45565 (95% paired-cluster interval, −1.15857 to 0.02367; Supplementary Fig. 13). Nineteen network means favoured the ensemble forecast and 21 favoured the nominal forecast. The shared point predictions had MSE 1.1781×10^{-4}, so no point-prediction improvement can be attributed to changing the forecast alone. The interval does not establish a stable forecast-score gain. The mean is also dominated by one network: its score difference of −11.67 accounts for about 64% of the summed differences, and the mean of the other 39 is −0.17. In

aggregate, the nominal forecast risk (1.19×10^{-4}) was closer to the actual MSE (1.18×10^{-4}) than the ensemble forecast risk (1.92×10^{-4}), so this study does not show that the ensemble forecast is better calibrated than the nominal forecast. The exploratory aggregate hint from the first study (ensemble ratio 1.10 against nominal 1.62) therefore did not replicate. No fit failures, fallbacks or forecast-floor clipping occurred.

A secondary hidden-bias-shift comparison favoured the ensemble forecast by −1.3583 score units (pointwise 95% interval, −2.2143 to −0.6345). Both mean risk forecasts nevertheless remained below actual MSE in that condition. This result describes a relative forecast improvement in one secondary setting; it does not show that the hidden shift was learned or that uncertainty was fully calibrated.

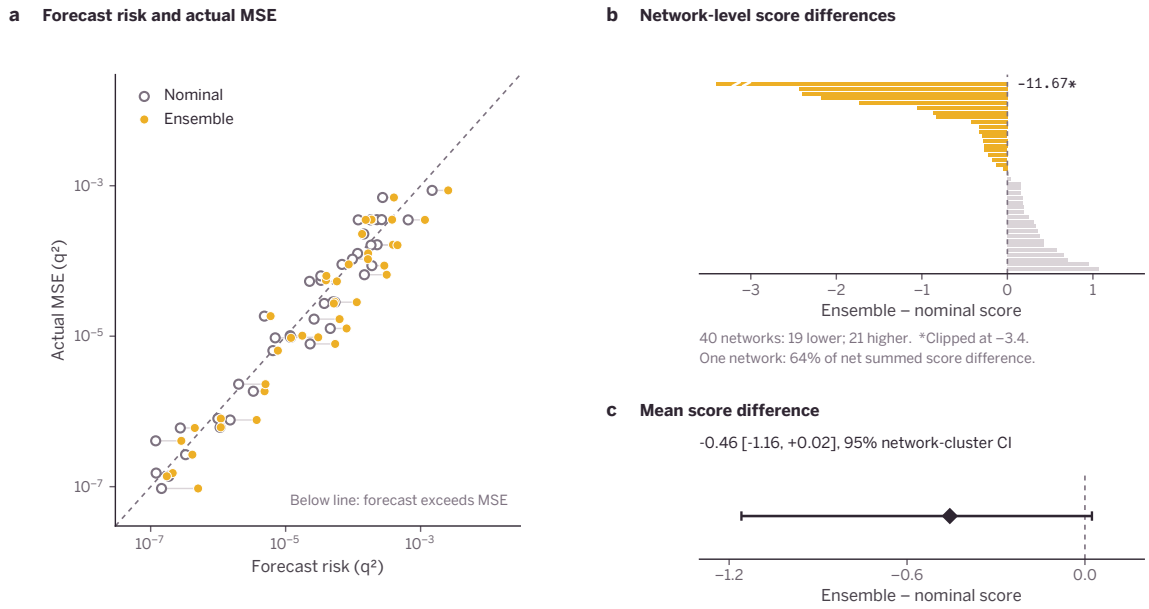


Fig. S13 | Risk forecasts are evaluated on identical predictions. The independent follow-up uses the nominal scalar recording policy, broad matched uncertainty, excluded endpoints and M=3 in 40 new networks. **a,** Nominal (open) and ensemble (filled) mean forecast risks are paired with the same actual mean response MSE for each network; horizontal segments link the two forecasts. Axes have identical logarithmic ranges and a dashed identity line; points below the line have a forecast that exceeds the actual error. This network-aggregate comparison is descriptive, not a stand-alone calibration test. **b,** All 40 network-level score differences, sorted; 19 favour the ensemble forecast (amber) and 21 the nominal forecast (grey). One bar (−11.7) is clipped at the axis limit and labelled. **c,** The mean score difference with the prespecified 95% paired network-cluster interval. Panels b and c use different horizontal scales because individual differences and uncertainty about their mean are distinct statistical objects. Scores $S(v,e)=\log(v)+e^2/v$ are computed for individual target/truth/noise cases and then averaged within network; they are not scores of the aggregate coordinates in a. Negative differences favour the ensemble. The score evaluates an MSE functional, not a complete predictive distribution.

Independent confirmation narrows the benefit suggested by the 100-unit pilot

An exploratory extension used eight independent 100-unit rank-two tanh networks from a published mathematical family [9], with adapted calibration inputs, measurement noise and inference. It suggested lower response error from prior averaging in several settings, including a −13.56% shared-objective change at five recordings with endpoints excluded (95% CI −28.56% to −4.78%). We therefore fixed that contrast as the primary comparison in 40 new networks before inspecting their outcomes. The original pilot and new confirmation remain separate sampling stages.

In the confirmation, prior averaging changed shared-objective mean MSE by +7.95% (95% CI −23.59% to +20.95%; 17 of 40 networks favoured the ensemble). Thus, the pilot improvement was not established in the larger independent sample. At ten recordings, large fitting errors made the aggregate contrast especially sensitive to individual networks. The complete objective, budget and pool results are retained in the supplement.

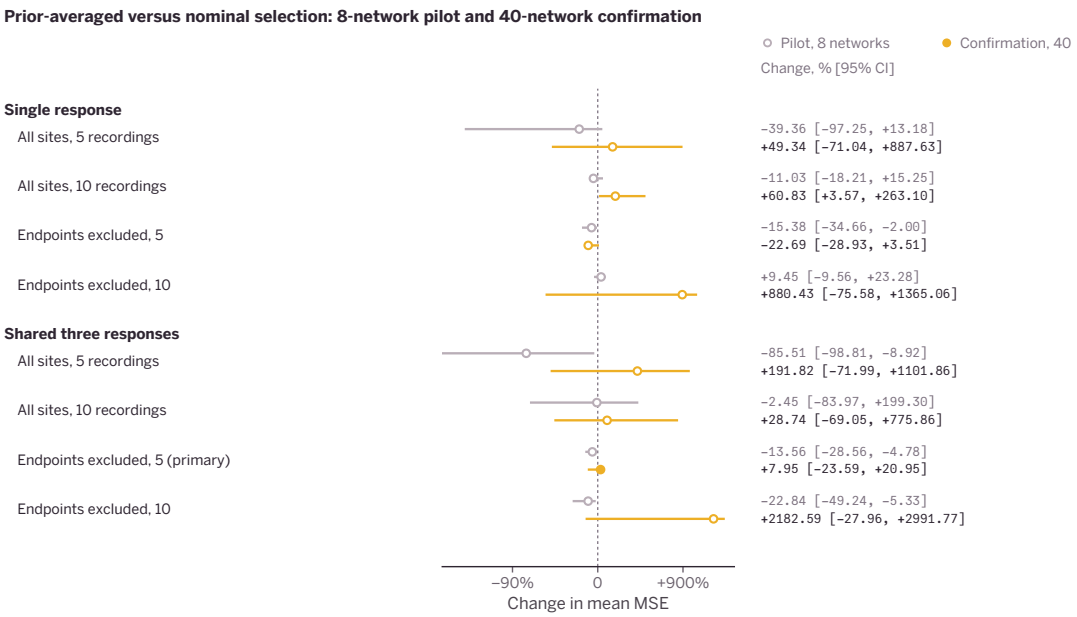


Fig. S14 | Pilot and independent confirmation in the 100-unit family. The eight-network pilot (open grey) and the 40-network confirmation (gold) are plotted separately as changes in aggregate mean MSE for prior-averaged relative to nominal selection, paired within network, with 95% network-bootstrap intervals on a logarithmic ratio axis. The prespecified confirmation contrast is the shared objective with endpoints excluded and five recordings (filled marker); other conditions are exploratory. Changing from the 12-unit model to this family also changes dynamics and parameterisation.

Uncertainty changes decisions while fitting remains a separate source of variation

Within the same 40 confirmation networks, prior and truth standard deviations were varied together over 0.3, 0.6 and 0.9, with common standardised draws. The mean correlation between nominal and ensemble candidate rankings at the first selection step fell from 0.944 to 0.687 to 0.600. All 40 networks showed a decrease between the narrowest and widest distributions. At five recordings, the average fraction of shared selected sites fell from 0.735 to 0.580 to 0.555. Thus uncertainty substantially changes recording choices in this family, while the corresponding response-error comparisons remain variable. Component controls separating target sensitivities and observation information did not establish one consistently dominant source of error reduction.

We also revisited the primary five-recording comparison using one common fitting rule: the original start, two nearby starts and two prior draws, retaining the smallest MAP objective. The selection rule used calibration data and the prior, with no access to response truth. Across the same 40 networks, ensemble versus nominal selection changed MSE by -49.40% (95% CI -70.90% to +3.65%). For nominal and ensemble selection separately, changes from the original fit were -3.07% (95% CI -13.14% to -0.27%) and -54.56% (95% CI -76.12% to +2.26%). One network contributed 80.7% of the absolute between-selector difference after refitting; three nominal fits still had objectives above the objective at the generating parameters. All 320 retained best fits terminated successfully. These are retrospective estimator-sensitivity results, reported alongside the frozen confirmation rather than replacing it.

Endpoint recording improved functional predictions much more clearly than it recovered individual gains. In the first cohort, mean stimulation-gain MSE changed from 0.661 to 0.579 and mean readout-gain MSE from 0.787 to 0.760 between the neither and both allocations. The validation cohort gave corresponding changes from 0.700 to 0.287 and from 0.743 to 0.507. A nominal local calculation predicted marginal variance reductions from approximately 91% of prior variance to 1–2%, which should be distinguished from these nonlinear parameter-estimation errors. In a separate oracle diagnostic on the first cohort, replacing fitted stimulation and readout gains by their true values, while leaving all other fitted parameters fixed, changed the neither-allocation MSE by -45.45% (95% CI -91.95% to

Table S2 | Earlier-study computational accounting

Stage	Independent J	Truth × noise × readouts per J	Evaluation records	Actual MAP calls	Core runtime (s)
Development	8	2 × 2 × 3	19,200	5,696	25.4
Frozen recording evaluation	40	3 × 3 × 3	129,600	38,610	356.0
Independent forecast study	40 new	3 × 3 × 3	6,480	3,006	17.5
Exploratory 100-unit family	8 new	2 × 2 × 3	3,072	1,436	693.1

Connectivity matrices are the independent sampling units. Truth, measurement noise and readout repeats are nested within J. Timings are core experiment wall times on the study Mac and are not cross-platform performance guarantees.

+36.98%). These initial diagnostics motivate the next experiment: endpoint observations can change inferred recurrent dynamics and parameter compensation, so the relevant question is how far response predictions can vary among comparably fitted models.

The improvement was specific to the intended response. Under different held-out calibration-style inputs, whole-network activity MSE changed by +5.36% (95% CI -12.00% to +30.00%) in the first cohort and +8.05% (95% CI -23.95% to +39.06%) in the validation cohort, despite the large response-error reductions. In an exact descriptive decomposition of response error into a readout-gain term and a dynamic-response term, their negative cross term offset 93.8% and 83.7% of the sum of their separate squared terms under joint endpoint recording, compared with 23.5% and 2.1% under neither-end recording. This supports studying the predictive combinations constrained by calibration, rather than interpreting parameter accuracy alone as the mechanism.

Approximation diagnostics constrain the interpretation

Development-only diagnostics measured local Taylor remainders and selection overlap. Mean calibration remainders relative to parameter-induced activity changes were approximately 0.179 under matched uncertainty, 0.334 under broad uncertainty and 0.252 under hidden bias shift. Broader uncertainty changes signal scale and parameter variability as well as nonlinear behaviour, so these differences do not isolate a curvature effect. Local and ensemble sets frequently overlapped. When sets were identical, the fitted data and response predictions were identical by construction, limiting any decision benefit of a changed risk value in those cases.

The matched and hidden-shift conditions used the same public prior and therefore produced exactly the same prospective designs, despite different true responses. This is an information boundary: without a preliminary measurement, the selector cannot infer an undisclosed physiological shift. In the first frozen evaluation, approximate 95% local MAP response intervals for the nominal scalar estimator covered the truth with proportions 0.930, 0.919 and 0.793 for matched, broad and shifted conditions, respectively, at excluded endpoints and M=3. These diagnostics identify limitations of the current information and approximation contract; they do not establish a single biological mechanism for the observed errors.

Earlier-study computational accounting

Core runtime is recorded separately from environment preparation, compilation, literature review and figure production. Dataset reuse avoids repeated optimisation of identical selected observations; evaluation rows are not independent sample counts.

The independent 100-unit confirmation added 40 networks, 15,360 scalar response records and 7,112 actual MAP calls. The matched-width diagnostics reused those networks (11,520 records; 3,296 calls). The focused endpoint experiment reused the same 40 networks (1,920 joint-response fit records; 1,916 calls), and independent endpoint validation added 40 new networks (1,920 records; 1,908 calls). The common five-start fitting analysis reused 320 datasets and made 1,600 fit calls. Two endpoint-validation calls reached the evaluation cap with finite outputs, which were retained. Thirty-seven alternative starts in the fitting follow-up reached the cap; all selected best fits terminated successfully. Before the final

Table S3 | Independent fifth-recording follow-up

Outcome	Comparison	Change (%)	95% interval (%)
Response MSE, primary	Response-directed / random fifth	+0.24	-41.36 to +74.08
Response MSE, secondary	Response-directed fifth / four endpoints	-30.08	-52.31 to -16.17
Response MSE, secondary	Random fifth / four endpoints	-30.25	-56.29 to -5.98
Response MSE, secondary	Combination-directed / random fifth	+13.50	-37.19 to +89.23
Response MSE, secondary	Response-directed / combination-directed fifth	-11.68	-22.22 to +5.44
Combination MSE, secondary	Combination-directed / random fifth	-31.07	-42.35 to -19.81

Changes are ratios of aggregate mean errors, expressed as percentages. All five-site policies include the stimulation site and three readouts. Combination MSE uses the four orthonormal internal coordinates. The four-endpoint comparator uses one fewer recording.

follow-ups, the 100-unit evidence comprised eight pilot networks, forty criterion-confirmation networks and forty additional endpoint-validation networks. The complete cohort accounting appears in Methods.

Independent fifth-recording comparisons

The eighty-network follow-up used one physiological draw and three paired noise repeats per network. Errors were averaged over the three targets and noise repeats within each network before comparing allocations. The response-directed versus random fifth-site contrast was the single prespecified primary comparison. Remaining intervals are descriptive pointwise 95% paired-network bootstrap intervals. All selected fits converged and passed the generating-parameter feasible-cost check. Re-evaluating target responses with twice the integration resolution changed the primary effect by 0.003 percentage points.

At a shared reference fit to the four endpoints, combination-directed selection reduced local internal variance by 26.50% relative to a random fifth site (95% CI -33.42% to -20.42%). The corresponding local response-variance change was -5.00% (95% CI -13.08% to -1.27%). This smaller contraction in the response quantity accompanies the stronger recovery of internal combinations. Central finite responses along internal covariance modes departed from the local sensitivities by a median of 1.4% at standardized length 0.1 and 11.7% at length 0.3.

The finite-error decomposition reconstructs response error from endpoint, internal-combination and remaining parameter contributions, including their cross terms. The repeated-noise records also support a decomposition into noise-averaged prediction bias and noise variance. Per-network contributions, uncertainty blocks and repeated-noise predictions are available in `round6/analysis` and `round6/runs`.